Adverse Selection Among Early Adopters and Unraveling Innovation

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Abstract

I study "selection markets" where consumers differ in willingness-to-pay and the costs they impose on sellers. I characterize when adverse selection is stronger in immature markets: early adopters willing to buy despite lower initial valuations impose the highest costs. This generates a dynamic free-rider problem that can cause markets to unravel before reaching maturity, even when the mature equilibrium would be efficient. Market power can mitigate this by enabling intertemporal cross-subsidization. Exploiting age-based differences in private information, I find that adverse selection among early adopters in the deferred income annuity market declines as the market matures.

Keywords: Adverse selection, Insurance, Annuities; JEL Codes: D82, G22, D14

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1. Introduction

Selection markets—where consumers differ in the costs they impose on firms—are ubiquitous across annuities, insurance, credit, securities, used goods, and even labor markets.[1] These markets often exhibit low utilization, which empirical work typically attributes to low consumer demand.[2] However, this explanation overlooks a key puzzle: suppliers' failure to offer more attractive products despite unmet demand. I show how adverse selection in immature markets exacerbates initial low demand and causes unraveling. Even markets that would be fully efficient at maturity fail to develop when early adopters are more adversely selected.

This paper's central insight is that adverse selection in immature markets creates a novel dynamic free-rider problem. When early adopters are more adversely selected, markets may fail to mature because initial entrants bear the costs of adverse selection while subsequent entrants compete away long-run rents: preventing the cross-subsidization of initial losses. Consequently, initial adverse selection can cause permanent unraveling even when the mature market would be efficient.

Whether early adopters are in fact more adversely selected is central to this argument—a theoretically ambiguous question. I formalize sufficient conditions with theory and provide supporting evidence from the maturing deferred income annuity market. In immature markets, potential consumers have less confidence in products and derive lower benefits from adoption. As demand falls, selection patterns change: early adopters willing to buy despite lower initial valuations tend to be those imposing the highest costs on firms, so adverse selection is higher in immature markets under economically motivated conditions on the joint distribution of costs and willingness-to-pay.

Using hand-collected annuity price data, I exploit age-based differences in available private information to separate adverse selection's role from other pricing dynamics. I validate the key identifying assumption—that private information increases with purchase age—using subjective mortality beliefs from the Health and Retirement Study, showing that older respondents' beliefs are significantly more predictive of realized mortality even conditional on health observables. I find economically and statistically significant decreases in adverse selection as the market matures. Groups with more

[1] Einav et al. (2021) provide an overview.

[2] See Lockwood (2012) and Pashchenko (2013) for annuities, Ameriks et al. (2016) and Lockwood (2018) in long-term care insurance, and Nakajima and Telyukova (2017) and Cocco and Lopes (2020) in reverse mortgage markets. Alternatively see Einav et al. (2010) for results emphasizing adverse selection.

private information experience larger equilibrium price reductions, consistent with theory predictions. Taken together, this provides a *joint* explanation for three empirical regularities: low take-up of existing products, sluggish demand for new products, and market inactivity despite substantial unmet demand. For example, Zinman (2014) highlights missing rungs in the consumer lending ladder, Ameriks et al. (2016) documents unmet demand for better long-term care insurance, and Cocco and Lopes (2015) establish demand for alternative reverse mortgage products. Michaud and Amour (2023) document potential demand for bundled risk management products combining existing insurance. Yet take-up remains very low: Nakajima and Telyukova (2017) and Webb (2011) document poor adoption of novel reverse mortgage and annuity products, respectively. Similarly, despite being available since 1954, variable annuities (bundling tax-favorable saving, insurance, and indexed options) remained a small market until the early 1990s (Brown and Poterba 2006).[3]

The paper makes three important contributions. First, I develop general comparative statics results for selection markets under monotonicity assumptions, microfounding the payoffs that generate the dynamic free-rider problem. Since monotonicity is not guaranteed with multidimensional heterogeneity (Fang and Wu 2018), I characterize sufficient, and economically interpretable, conditions using copulas governing the joint distribution of costs and willingness-to-pay. These conditions accommodate commonly used empirical specifications, extend the graphical framework of Einav and Finkelstein (2023) to cross-market and dynamic comparisons by characterizing demand and cost curve shifts, and deliver a partial ordering of equilibrium prices, quantities, selection, and welfare in terms of model primitives.

Second, I outline the novel free-rider problem and its implications in an intuitive framework. A key insight is that even would-be efficient markets can unravel during immaturity—a new, dynamic source of market failure building on Akerlof (1970). This challenges standard intuitions about market power and adverse selection. Static analyses, such as Starc (2014) and Mahoney and Weyl (2017), highlight that adverse selection can mitigate market power's negative impacts by constraining rent extraction, though market power exacerbates under-supply. In a dynamic setting, however, market power can alleviate these dynamic inefficiencies by allowing monopolists to cross-subsidize early losses with longer-run gains. This points toward patent protection

[3] Concern about missing insurance innovation dates back over half a century: Rudelius and Wood (1970) identify heterogeneous adoption of life insurance innovations, Dorfman (1971) provides evidence of research activity and barriers to innovation, and Murray (1976) surveys insurance companies to identify innovations.

or temporary regulatory barriers as welfare-improving rather than the standard prescription of restricting market power.

Third, although testing hypothetical dynamics in nonexistent markets is impossible, I show the model predictions are empirically relevant using the market for deferred income annuities. I use a novel identification strategy exploiting age-based differences in the potential for private information to test for changes in adverse selection as the market matures. Adverse selection declines as the market matures and does so more strongly for older purchasers, who I show possess the most private information.

All three contributions build on a standard insurance demand model with multidimensional heterogeneity. Consumers are heterogeneous across costs, risk premia (which determine insurance surplus), and other dimensions capturing frictions or departures from rationality. I consider adverse selection settings where consumers' willingness-to-pay correlates positively with their costs to insurers.[4] To study unraveling in novel markets, I distinguish mature markets (well understood by consumers) from immature markets (new and unfamiliar), allowing consumers' willingness-to-pay to vary with market maturity.

These differences in components of willingness-to-pay capture objective factors, such as counterparty risk (Briggs et al. 2023) or self-insurance patterns consumers cannot quickly adjust, and subjective factors including distrust of new products, inertia (Handel 2013), or learning dynamics (Israel 2005). They lower demand and increase consumer sorting on costs, increasing adverse selection in immature markets and generating equilibrium under-insurance with lower adoption and higher costs.

Related Literature

I show that a novel free-rider problem manifests in the dynamics of adversely selected markets. Importantly, this reverses standard intuition for the interaction of market power and adverse selection. At any point in time, adverse selection impacts firm's ability to exploit market power as discussed by Starc (2014) and Mahoney and Weyl (2017). Moreover, market power exacerbates the under-supply driven by adverse selection. Here, however, the presence of market power early in a product's life cycle can incentivize entry when would-be efficient markets fail to mature.

This insight connects closely to Kong et al. (2024), who demonstrate that excessive price competition arising from "cream-skimming" incentives in adversely selected markets can undermine entry incentives and lead to market unraveling. This shows

[4] Analogous results can be obtained for advantageous selection.

that market power can actually improve welfare in selection markets by preserving entry incentives. My analysis extends this fundamental insight to a dynamic setting, showing how the timing of competition matters crucially for market development. While Kong et al. focus on static equilibria where competition prevents profitable entry, I examine how the evolution of competition—with early entrants facing adverse selection but later entrants competing away rents—creates a dynamic free-rider problem that can prevent markets from ever reaching maturity. The key distinction is temporal: early entrants must bear the costs of market development (including higher adverse selection) while later entrants can free-ride on this investment by entering only after the market matures. Kong et al. similarly find a role for restricting competition, but the dynamic mechanism here points toward protecting early entrants' future rents, via patent protection or temporary regulatory barriers, so that intertemporal cross-subsidization remains feasible. The potential optimality of intertemporal cross-subsidization is related to, but different from, policies designed to mitigate selection (Geruso and Layton 2017, review these policies in health insurance markets).

To microfound these dynamic incentives, I develop comparative static tools providing non-parametric conditions for ordering partial unraveling in Akerlof markets. This characterization allows theoretical insights to map onto commonly used empirical specifications, such as the constant absolute risk aversion utility in Cohen and Einav (2007). Building on Akerlof (1970)'s analysis of market unraveling, Hendren (2013) provides extended no-trade theorems for Rothschild and Stiglitz (1976) markets, while Attar et al. (2021) establish necessary and sufficient conditions for no-trade theorems in adverse selection markets.[5] I extend non-entry results to cases where adverse selection declines over time, showing how these dynamics can lead to permanently missing markets even when static analysis predicts efficiency.

I study distributional changes in choice frictions similarly to Handel et al. (2019), providing comparative statics for non-local changes in both choice friction distributions and insurance value of contracts across markets. This builds a partial ordering of equilibrium prices, quantities, and welfare in terms of model primitives. Handel et al., however, study local reforms (including information interventions) that alter consumer's choice frictions and provide sufficient statistics formulas for their welfare impacts. As in Handel et al. and Spinnewijn (2017), I show how changes to model primitives lead to the rotations and shifts to cost and demand that are studied in

[5]A sizable literature studies endogenous contract terms under adverse selection (e.g. Rothschild and Stiglitz 1976; Handel et al. 2015; Veiga and Weyl 2016; Azevedo and Gottlieb 2017) or screening with intensive and extensive margins (Geruso et al. 2023), which is not this paper's focus.

other contexts (e.g. Starc 2014; Mahoney and Weyl 2017).

2. Adverse Selection in Mature and Immature Insurance Markets

I develop a canonical model of insurance demand and use it to study how market maturity affects adverse selection. I present the framework under competitive pricing and then introduce differences between mature and immature markets that generate stronger adverse selection among early adopters (Sections 2.1-2.4). I subsequently, extend it to market power and product differentiation (Section 2.5), and embed the resulting payoffs in a two-period dynamic entry game (Section 3)

2.1. Demand, Supply, and Adverse Selection

I consider an Akerlof (1970) model of insurance in the style of Einav et al. (2010) and Einav and Finkelstein (2011). A single perfectly competitive and risk-neutral insurer sells insurance to a unit continuum of heterogeneous individuals at a common price p. Contract terms are exogenous and common across all individuals.

Demand. Individuals are heterogeneous across risk type π_i which determines the expected cost to the insurer; their risk premium r_i; and potential demand frictions ε_i. These are private information for individual i.[6] An individual buys insurance if and only if their *subjective* willingness-to-pay, defined as

$$\omega_i \equiv u(\pi_i, r_i, \varepsilon_i) = u(\underbrace{\pi_i + r_i}_{\text{insurance value}} + \varepsilon_i), \tag{1}$$

for any strictly increasing utility function $u(\cdot)$, exceeds the price of the insurance. In addition to their risk type, individual i's risk premium, r_i, determines the surplus in the insurance market.

Following Spinnewijn (2017), individual choices depend on additively separable demand frictions, ε_i, which can distort insurance demand. This agnostic approach nests important features including borrowing constraints, inertia, subjective beliefs, and bounded rationality. These frictions are heterogeneous and can lead individuals to

[6] Conditioning on observable characteristics correlated with π_i may be illegal. Nevertheless, this can be interpreted as a model of insurance contracts in each partition of the observable characteristic space. This rules out screening contracts or changes to the screen – this may be particularly important in immature markets.

overvalue ($\varepsilon > 0$) or undervalue ($\varepsilon < 0$) the insurance contract. Since risk premia and frictions have identical effects on consumer choices, I focus on perceived risk premia net of behavioral influences $\tilde{r}_i \equiv r_i + \varepsilon_i$, for the majority of the analysis.

Individual demand is given by $\mathbb{1}[\omega > p]$ which yields aggregate demand

$$D(p) = 1 - F_\omega(p) = E\left[\mathbb{1}[\omega > p]\right], \tag{2}$$

where F_x denotes the CDF of variable x with mean, μ_x, and variance, σ_x^2.

Supply. The cost of providing insurance at price p is determined by the risk types of individuals purchasing insurance at that price.[7] Consequently, *average* and *marginal* costs are given by:

$$AC(p) = E(\pi|\omega \geq p) \tag{3}$$

$$MC(p) = E(\pi|\omega = p)\,. \tag{4}$$

When willingness-to-pay, ω, increases in insurer cost, π, there is adverse selection; marginal costs increase in price and the MC curve is downward sloping.

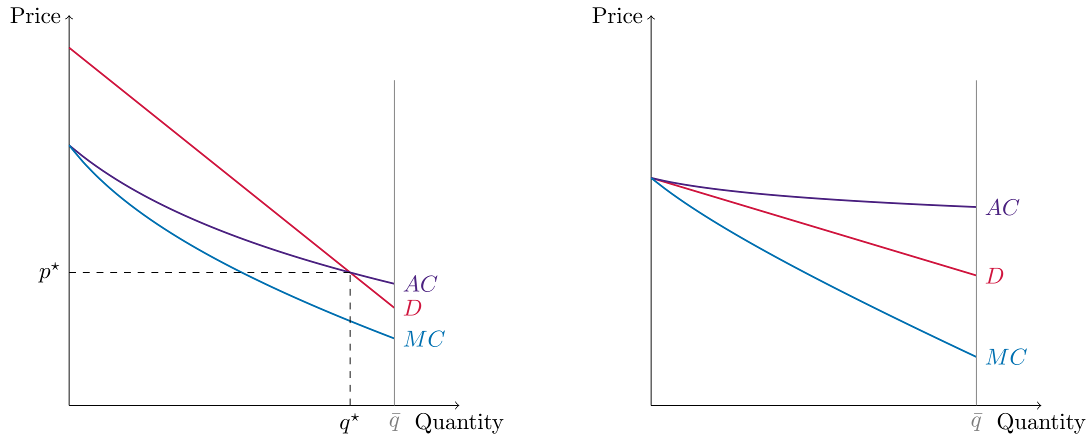
Equilibrium. Assuming continuous, monotonic demand and cost curves ensures a unique equilibrium if it exists. The equilibrium price is given by the break even price that pools expected costs

$$p^\star : p^\star = AC(p^\star)\,. \tag{5}$$

Consequently, the equilibrium allocation may differ from an efficient allocation. An efficient allocation insures everyone whose willingness-to-pay exceeds their expected cost ($\omega_i > \pi_i$), those for whom the demand curve lies above the marginal cost curve (Panel A, Figure 1).

Furthermore, a positive equilibrium price does not necessarily exist. The insurance market unravels (Akerlof 1970) when individuals are not willing to purchase insurance at the pooled cost of higher-demand individuals. This occurs when the demand curve lies entirely below the average cost curve (Panel B, Figure 1 and Hendren 2013).

[7] I abstract from administrative loadings or fixed costs which don't affect the main result.



(A) Textbook Adverse Selection

(B) Textbook Unraveling

FIGURE 1. Equilibrium in the Insurance Market Without Demand Frictions

2.2. Mature and Immature Markets

I use superscripts *MM* and *IM* to denote the mature and immature markets and the relation $X \preceq Y$ to denote first order stochastic dominance. To make markets comparable, I hold the distribution of potential and realized risks fixed across mature and immature markets so that the product and risk pools are identical.[8]

ASSUMPTION 1. *Risk types, π_i, have identical marginal distributions in mature and immature markets.*

I then assume willingness-to-pay is depressed in immature markets, thus demand contracts. This can be driven by objective risk premia, r, or from demand frictions, ε. I begin by motivating differences in the objective component of net premia, $\tilde{r}$.

Three forces generate objective differences between risk premia in immature versus mature markets. First, portfolios may not yet be optimized to take advantage of a new product (e.g., savings used for self-insurance crowding out formal insurance), so the benefits of insurance rise as other choices adjust. Second, there may be greater objective uncertainty about coverage, hassle costs, or reclassification risk; in health insurance, for example, coverage of new drugs/procedures is uncertain and heterogeneous, which depresses premia. This uncertainty is likely larger for new types of insurance and lowers the risk premia. Third, counterparty risk may be higher when insurers collect premia now but only begin paying out far in the future (e.g., life

[8]Thus neither firms nor individuals can alter the risk distribution in mature markets.

insurance, deferred annuities, long-term care), distorting insurers' portfolio choice (Thompson 2010, models endogenous counterparty risk) and raising the likelihood of default or non-payment—again lowering risk premia.

Alternatively, lower demand may reflect more negative demand frictions in immature markets: status quo bias, distrust of new products, inertia, lack of awareness (Boyer et al. 2020), or narrow consideration sets. Frictions that dissipate with learning or peer experience (Chetty et al. 2013; Duflo and Saez 2003) fit the same representation. In short, immature markets have *more negative* demand frictions than mature markets, where more consumers have prior exposure (peers, advertising, past purchase).

Assumption 2. *Willingness-to-pay is on average larger and demand expands in mature markets:* $\omega^{MM} \succeq \omega^{IM}$

This formalizes how shifts in risk premia and demand frictions affect willingness-to-pay. Instead of demand frictions shifting systematically, consumers may have noisier beliefs about new products, reflected in greater dispersion of demand frictions capturing unbiased but diffuse priors. Assumption 2 is weaker than requiring stochastic dominance of objective risk premia, r, or demand frictions, ε. It only needs to be satisfied *on net*. Unlike Spinnewijn (2017), this does not require consumers are unbiased on average; systematic optimism or pessimism is consistent with Assumption 2 (see, e.g. Brown et al. 2021; O'Dea and Sturrock 2023).

Assumptions 1 and 2 restrict how marginal distributions differ between mature and immature markets. Yet, they do not capture how the sorting of consumers may change as markets mature. A natural primitive is that, in immature markets, cost-independent sources of willingness-to-pay heterogeneity are compressed. This weakens the dependence between π_i and ω_i.[9] I formalize this as a concordance ordering on the copula between π_i and ω_i:

Assumption 3. *Resorting is such that copulas C, describing dependence between π_i and ω_i, satisfy the following concordance ordering (Yanagimoto and Okamoto 1969; Tchen 1980; Scarsini 1984):* $C^{IM}(u,v) \geq C^{MM}(u,v)\ \forall\, u,v \in [0,1]$

This orders joint distributions *between* markets by implied dependence. Costs and willingness-to-pay have larger concordance—their joint density is more concentrated

[9] To illustrate, decompose $\tilde{r}_i = h(\pi_i) + \upsilon_i$ with $\upsilon_i \perp \pi_i$. If $\mathrm{Var}(\upsilon^{IM}) \leq \mathrm{Var}(\upsilon^{MM})$, the concordance ordering follows from the result that adding independent noise weakens concordance (Tchen 1980). Under joint normality with constant willingness-to-pay variance, this additionally requires the cost-correlated component of risk premia to decline with maturity—a plausible consequence of the same process, but one the general non-parametric results do not require.

around a monotone graph.[10] Intuitively, the highest willingness-to-pay types may impose the highest coverage costs, making early adopters uniformly more costly on average and changing how insurance purchase signals cost to insurers.

When consumers must expend resources to learn about a new contract or integrate it into an existing financial plan, those with the strongest insurance motive—high-π types—are disproportionately likely to adopt early, so the resulting pool of buyers is tightly sorted on insurer cost. As the market matures, adoption extends to consumers whose willingness-to-pay reflects portfolio complementarities, long-term planning, peer experience, or product-specific preferences rather than high expected cost alone. The assumption is intended to capture settings in which immaturity disproportionately suppresses demand among lower-cost consumers, so that early adopters are more selected on cost. The empirical evidence from deferred income annuities is consistent with this mechanism, but the framework does not require that it hold in *every* new market.

2.3. Equilibrium in Immature Markets

I use the graphical framework of Einav et al. (2010) and Einav and Finkelstein (2011) to characterize equilibrium differences between mature and immature markets. For each result, I provide closed-form proofs under joint normality; Section 2.4 gives general sufficient conditions for monotonicity and within-market results under arbitrary marginals (distinct from resorting in Assumption 3), with full proofs of general results in Appendix A.3.[11]

Assumption 4 (Joint Normality). *π_i and $\tilde{r}_i$ are jointly normally distributed. Additionally, the variance of willingness-to-pay is constant, $\sigma_\omega^{IM} = \sigma_\omega^{MM}$, and correlation between risk type and willingness-to-pay is weaker in mature markets, $corr(\pi, \omega^{IM}) \geq corr(\pi, \omega^{MM}) > 0$.*

Under joint normality, π and $\tilde{r}$ are multivariate normal with correlation $\rho_{\pi,\tilde{r}}$, and the joint distribution of cost type and willingness-to-pay is given by

$$\begin{pmatrix} \pi_i \\ \omega_i \end{pmatrix} \sim N\left[\begin{pmatrix} \mu_\pi \\ \mu_\omega \end{pmatrix}, \begin{pmatrix} \sigma_\pi^2 & \rho\sigma_\pi\sigma_\omega \\ \rho\sigma_\pi\sigma_\omega & \sigma_\omega^2 \end{pmatrix}\right],$$

[10] See Appendix A.3 for a construction from assumptions on π_i and $\tilde{r}_i$ directly.

[11] Monotonicity of demand and cost curves does not require choice frictions are independent of other characteristics (Solomon 2023 finds evidence in favor of dependence). Consequently, choice frictions are a reduced form compatible with a range of behavioral microfoundations including rational inattention models and correlated mistakes affecting both risk premia and purchase decisions (Leive et al. 2022).

where $\rho = \frac{\sigma_\pi + \rho_{\pi,\tilde{r}}\sigma_{\tilde{r}}}{\sigma_\omega} > 0$. The marginal cost curve is

$$MC(p) = E(\pi|\omega = p) = \mu_\pi + \rho\frac{\sigma_\pi}{\sigma_\omega}(p - \mu_\omega), \tag{6}$$

which is increasing in p when $\rho_{\pi,\tilde{r}}$ (and thus ρ) is positive. Average costs are

$$AC(p) = E(\pi|\omega \geq p) = \mu_\pi + \rho\sigma_\pi\frac{\phi(\frac{p-\mu_\omega}{\sigma_\omega})}{1 - \Phi(\frac{p-\mu_\omega}{\sigma_\omega})}, \tag{7}$$

an endogenously truncated normal distribution, increasing in p for positive $\rho_{\pi,\tilde{r}}$.[12]

Demand Contraction. The implication of Assumption 2 for demand is immediate: at any given price, average willingness-to-pay is higher in mature markets, so more individuals purchase insurance. Panel A of Figure 2 depicts this *level effect* on demand: $D^{IM}(p) \leq D^{MM}(p) \ \forall p$.

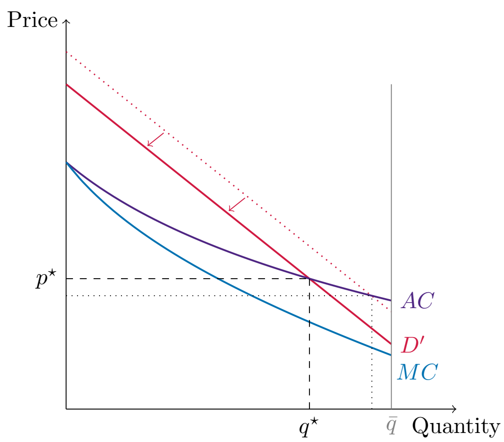
This level effect holds average and marginal costs fixed at any given quantity. Although households require lower prices to induce purchase, willingness-to-pay ordering is unchanged and the identity of marginal (and inframarginal) buyers at a given quantity is the same, thus equilibrium price rises while quantity falls (Panel A).

Cost Expansion. A pure level effect can raise average costs at a given price because it changes equilibrium quantity, but a change in sorting is required to shift cost curves *conditional on quantity*. When selection on cost increases, insurers face steeper marginal costs and higher average costs.

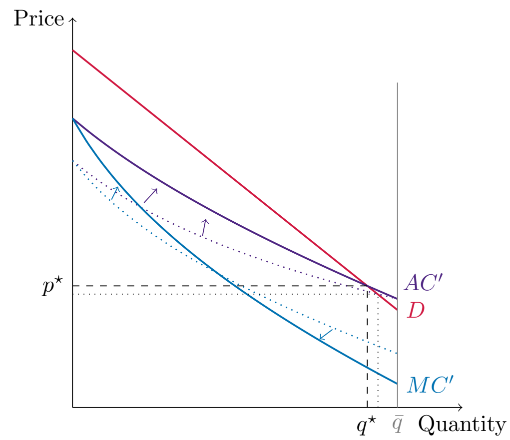
Order individuals by willingness-to-pay. Without re-sorting, this ordering is identical in mature and immature markets. Thus if q individuals buy insurance, the marginal consumer is the same and imposes identical expected costs on the insurer. With re-sorting, identities change: increased sorting on cost in immature markets implies that purchasing insurance signals higher average costs than in maturity.

Under normality, the decomposition of the difference in average costs makes this

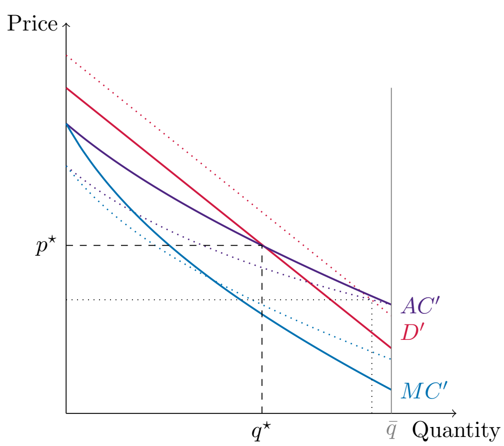
[12] Assumption 4 rules out changes in the dispersion of willingness-to-pay across mature and immature markets. Under this condition, changes to the primitives take the form of location shifts for the marginal distributions. This is weaker than the stochastic ordering of normals as it does not impose a common variance across mature and immature markets for π_i and $\tilde{r}_i$.



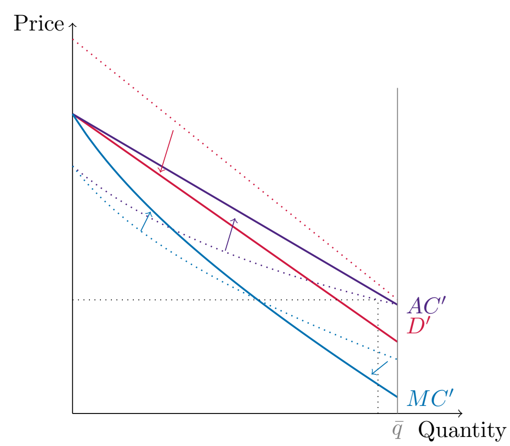
(A) Demand Contraction (Level Effect)



(B) Increased Adverse Selection (Sorting Effect)



(C) Under-insurance in the Short Run



(D) Unraveling in the Short Run

FIGURE 2. The Long and Short Run of Insurance Markets

precise. Comparing across markets gives, $\forall\, p$:

$$\frac{AC^{IM}(p) - AC^{MM}(p)}{\sigma_\pi \rho^{IM}} = \underbrace{\left(\frac{\phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})}{1-\Phi(\frac{p-\mu_\omega^{IM}}{\sigma_\omega})} - \frac{\phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})}{1-\Phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})} \right)}_{\text{level effect}\geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right) \frac{\phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})}{1-\Phi(\frac{p-\mu_\omega^{MM}}{\sigma_\omega})}}_{\text{re-sorting effect}\geq 0} \geq 0 \tag{8}$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution, ensuring non-negativity. Both effects work in the same direction—raising average costs at a given price—as additional selection on costs among inframarginal consumers dominates the ambiguous effect on the marginal consumer.

The corresponding decomposition for marginal costs is:

$$MC^{IM}(p) - MC^{MM}(p) = \frac{\sigma_\pi}{\sigma_\omega}\rho^{IM}\left[\underbrace{(\mu_\omega^{MM} - \mu_\omega^{IM})}_{\text{level effect}\geq 0} + \underbrace{\left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right)(p - \mu_\omega^{MM})}_{\text{re-sorting effect}}\right]. \tag{9}$$

Absent re-sorting, the increase depends only on the contraction in demand. With re-sorting, the sign is ambiguous: re-sorting raises the difference at prices above $\mu_\omega^{M}M$ but lowers it below, as the most adversely selected consumers have already been pulled out of the low-price pool.

Panel B of Figure 2 shows the outward expansion of the average cost curve in immature markets.[13] Even holding the level of demand fixed, increased adverse selection among inframarginal consumers raises expected costs. Insurers set higher prices to cover these costs, despite some lower-cost buyers exiting.[14]

Substituting inverse demand into (7) gives the decomposition at a given quantity:

$$AC^{IM}(q) - AC^{MM}(q) = \underbrace{\rho^{IM}\sigma_\pi\left(1 - \frac{\rho^{MM}}{\rho^{IM}}\right)\frac{\phi(\Phi^{-1}(1-q))}{q}}_{\text{re-sorting effect}} \geq 0 \quad \forall q. \tag{10}$$

At a given quantity, differences in average costs depend only on re-sorting. Marginal costs rotate around the median ($MC^{IM}(q) \geq MC^{MM}(q)$ for all $q \leq 1/2$ and vice versa), so the adverse selection wedge is higher at any quantity demanded: $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$.

Partial Unraveling. Demand contraction ($D^{IM}(p) \leq D^{MM}(p)$) together with cost expansion ($AC^{IM}(p) \geq AC^{MM}(p)$) jointly establish:

[13]This is a rotation around full insurance as in Mahoney and Weyl (2017). The difference in average costs in (10) is decreasing in quantity, reaching zero at full coverage: $AC^{IM}(1) = AC^{MM}(1) = \mu_\pi$. In contrast, Starc (2014) considers rotation around equilibrium demand.

[14]This can occur even if marginal costs in immature markets lie below demand which, absent choice frictions, would make full coverage efficient.

PROPOSITION 1 (Partial Unraveling). *If an equilibrium exists, the equilibrium price is higher in immature markets and fewer consumers are insured,* $q^{MM} \geq q^{IM}$ *and* $p^{MM} \leq p^{IM}$.

PROOF. I first establish underinsurance in equilibrium. Let q^{MM} be the equilibrium quantity in the mature market and let $\tilde{D}(q) = D^{-1}(q)$ denote inverse demand. Then evaluating at mature market equilibrium quantity

$$\tilde{D}^{IM}(q^{MM}) \leq \tilde{D}^{MM}(q^{MM}) = AC^{MM}(q^{MM}) \leq AC^{IM}(q^{MM}) \, , \tag{11}$$

where the first inequality is the (inverted) demand contraction and the second inequality is the cost-expansion. This establishes disequilibrium, $\tilde{D}^{IM}(q^{MM}) \leq AC^{IM}(q^{MM})$, which is strict unless the immature an mature markets are identical. Thus if inverse-demand and average cost intersect it must be at a lower quantity: $q^{IM} < q^{MM}$.

Using monotonicity of the average cost curve and that it is decreasing in q establishes

$$p^{IM} = AC^{IM}(q^{IM}) \geq AC^{IM}(q^{MM}) \geq AC^{MM}(q^{MM}) = p^{MM} \, . \tag{12}$$

Furthermore, if there exists a price $p^{IM} \neq p^{MM}$ that clears the immature market then differences across markets satisfy *strict* dominance. □

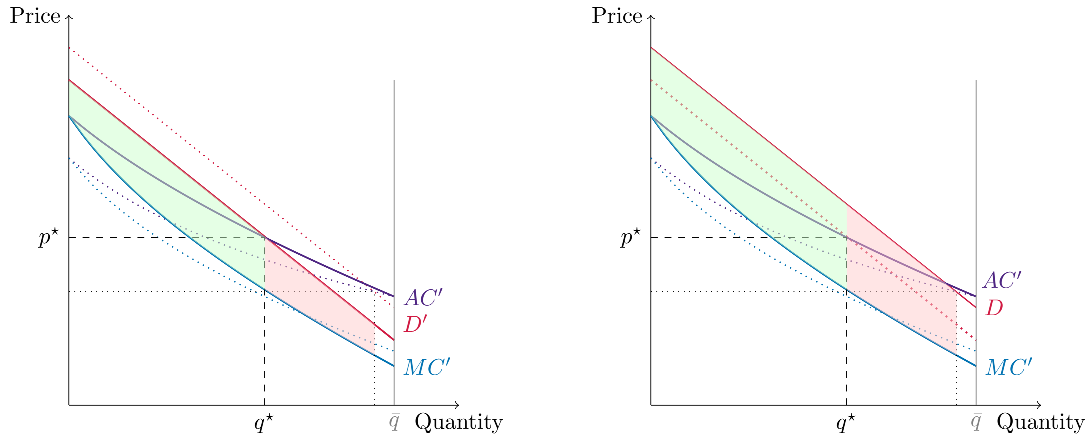
Panel C of Figure 2 illustrates the new equilibrium. Marginal costs lie below demand in both mature and immature markets, so full insurance is efficient in both. Nevertheless, relative to the mature market there is additional under-insurance due to both the level effect and increased adverse selection in the immature market. The equilibrium price increase and quantity reduction exceed either effect in isolation (Panels A and B).

Immaturity can also cause complete unraveling.

COROLLARY 1 (Complete Unraveling). *Would-be efficient mature markets can unravel before they reach maturity.*

This happens when the average cost curve lies above demand for all prices (Panel D of Figure 2). Thus, even a market that would be efficient if it could mature can unravel in its early stages. As Section 3 shows, this can lead to permanently missing markets.

Under-insurance ($q^{IM} < q^{MM}$) does not depend on whether lower willingness-to-pay is driven by risk premia or choice frictions. Computing welfare costs, however, requires a view on their relative importance. The next proposition evaluates welfare under two policy-relevant benchmarks, illustrated in Figure 3.



(A) Revealed Preference Benchmark (B) Paternalistic Benchmark

FIGURE 3. Welfare Losses in Immature Insurance Markets

PROPOSITION 2 (Efficiency Costs). *In the case of partial unraveling (Panel C), where* $q^{MM} \leq q^{IM}$*, consumer surplus in immature markets is lower at the immature market equilibrium than the mature market equilibrium* ($CS^{IM}(q^{IM}) \leq CS^{IM}(q^{MM})$)*. This is true irrespective of whether demand for insurance is evaluated at:*

i. Willingness-to-pay in immature markets (revealed preference benchmark)

ii. Willingness-to-pay in mature markets (paternalistic benchmark)

Thus, the equilibrium in immature markets is pareto dominated by the mature market.

PROOF. Firms earn zero profits in equilibrium, so consumer surplus is sufficient to order equilibria. Under the revealed preference benchmark, the difference in consumer surplus is

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q)\, dq \geq 0, \tag{13}$$

where $\tilde{D}(q) = D^{-1}(q)$ and non-negativity follows from $q^{MM} \geq q^{IM}$ (Proposition 1). Under the paternalistic benchmark,

$$CS^{IM}(q^{MM}) - CS^{IM}(q^{IM}) = \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{MM}(q) - \tilde{D}^{IM}(q)\, dq}_{\text{Paternalism } (\geq 0)} + \underbrace{\int_{q^{IM}}^{q^{MM}} \tilde{D}^{IM}(q) - MC^{IM}(q)\, dq}_{\text{Case (i)}(\geq 0)}, \tag{14}$$

where the paternalistic contribution is positive by demand contraction.[15] □

Under the revealed preference benchmark, the result requires that $q^{MM} \leq q^{IM}$, which holds when differences across markets reflect fundamentals rather than frictions inflating mature demand. The paternalistic benchmark (case ii) requires no such condition, since it evaluates immature surplus at mature preferences, where q^{MM} is by construction efficient.

2.4. General Sufficient Conditions

The preceding results are established under normality. With multidimensional heterogeneity, the critical assumption for adverse selection is the dependence between individual willingness-to-pay and costs (Fang and Wu 2018), which Assumption 3 pins down across markets. I use a copula approach to decouple this dependence structure from marginal distributions, allowing the analysis to extend flexibly beyond stylized examples.

The following proposition formalizes sufficient conditions for within-market monotonicity using the familiar monotone likelihood ratio property (Milgrom 1981):

PROPOSITION 3 (Adversely Selected Markets). *Let the density $f(\cdot)$ satisfy the Monotone Likelihood Ratio Property (MLRP), i.e. $f(\omega_1|\pi_1)/f(\omega_1|\pi_2) \geq f(\omega_2|\pi_1)/f(\omega_2|\pi_2)$ for any $\omega_1 \geq \omega_2$ and $\pi_1 \geq \pi_2$. Then:*

i. The market is adversely selected and marginal costs, $MC(p)$, are increasing in price.

ii. Average costs, $AC(p)$, are monotonically increasing in price.

iii. The adverse selection wedge, $AC(p) - MC(p)$, is positive.

The formal proof is given in Appendix A which shows that the MLRP property is sufficient, but not necessary. Intuitively, MLRP allows the insurer to infer higher costs from the signal of higher willingness-to-pay. Crucially, this does not depend on marginal distributions. For a bivariate distribution with copula $C(u, v)$ and copula density $c(u, v)$, MLRP holds if and only if:

$$\frac{\partial^2}{\partial u \partial v} \log c(u, v) \geq 0. \tag{15}$$

[15] It follows that consumer surplus in mature markets is generically larger than in immature markets.

This restricts the shape of the scale-invariant rank dependence *within* a market, formalizing that larger X values tend to be realized with larger Y values. It implies positive quadrant dependence and positive correlations and regression coefficients.[16]

Example.. *Consider the specification in Cohen and Einav (2007), where consumers have constant absolute risk aversion θ_i and face a loss m arriving with probability λ_i. Then $\tilde{r}_i = r_i = \frac{\ln\left(1-\lambda_i+\lambda_i e^{\theta_i m}\right)}{\theta_i} - \lambda_i m$, which is non-normal even when λ and θ are (log-)normally distributed. The copula conditions above, which impose no marginal distribution restrictions, extend the implication of the normality benchmark to such settings.*

Under these general conditions and Assumptions 1–3, demand contractions and cost expansions hold for arbitrary marginal distributions. Propositions 1 and 2 then follow immediately, as their proofs depend only on the ordering of monotonic demand and cost curves and not on a specific distribution. Formal statements and proofs are in Appendix A.3.

2.5. Differentiated Goods and Market Power

I now extend the model by relaxing product homogeneity and allow for market power. There are $j \in \{1, \ldots, J\}$ firms offering identical insurance contracts bundled with firm-specific differentiated attributes X_j. As in the equilibrium described above, firms maximize profits by setting a uniform price. Under Bertrand-Nash price competition, firm j's equilibrium pricing equation (replacing equation 5) becomes:

$$p_j^\star - AC(p_j^\star) = \frac{1}{\eta_{D,p_j}} - \left(AC(p_j^\star) - MC(p_j^\star)\right), \tag{16}$$

where the second term captures pricing pressure from adverse selection. Rearranging yields the standard monopolist pricing:

$$p_j^\star = \frac{1}{\eta_{D,p_j}} + MC(p_j^\star). \tag{17}$$

As in Mahoney and Weyl (2017), the impact of increased adverse selection depends on the quantities in the mature market:

[16] As dependence is measured in ranks, the cross-partial derivative captures more than the correlation between π and ω: it encodes how jointly heavy-tailed they are. Appendix A.2 derives sufficient conditions directly on the conditional family $\tilde{r}|\pi$ — namely, that $\tilde{r}|\pi$ satisfies MLRP in $(\tilde{r}, \pi)$ and each conditional density is log-concave — that imply MLRP for the induced joint distribution of (ω, π).

Lemma 1 (Under-Insurance with Product Differentiation). *Assume quantities in the mature market are small ($q_j^{MM} < \widetilde{q}$) and the elasticity advantage in mature markets is bounded so that $\eta_{D,p_j}^{MM}/\eta_{D,p_j}^{IM} \geq 1 - \eta_{D,p_j}^{MM}\left[MC^{IM}(q_j) - MC^{MM}(q_j)\right]$ for $q_j < \widetilde{q}$.*[17]

It follows that, if an equilibrium exists, equilibrium prices are higher and quantities lower in immature markets: $q_j^{MM} \geq q_j^{IM}$ and $p_j^{MM} \leq p_j^{IM}$.

This confirms that adverse selection reduces firms' incentive to supply differentiated products in immature markets, generating higher prices, lower quantities, reduced profits, and lower consumer surplus.[18]

Market power reduces but does not eliminate under-insurance in immature markets. The key question is whether a firm with market power can earn positive profits during immaturity — and if so, whether dynamic entry incentives can sustain the market through its early phase. I turn to this next.

3. Will Deep Pocketed Insurers Enter If Immature Markets Unravel?

Section 2.3 establishes that immature markets can suffer from worse adverse selection and may unravel even when the mature market would be efficient. I now ask whether dynamic incentives counteract these forces. Using a two-firm, two-period model, I show that adverse selection produces a novel form of dynamic free-rider problem that can prevent markets from ever reaching maturity.

In each period, firms choose whether to offer insurance taking contract terms as given. They observe entry decisions and post prices, competing à la Bertrand, so prices equal average costs if both firms enter. If only one firm enters, they exploit market power and equate marginal revenue with marginal cost. The market begins as immature and updates to mature in the second period if and only if at least one firm enters in the first period and sells positive quantity.[19]

The following result characterizes how market immaturity interacts with market forces and can endogenously create market incompleteness:

[17] This rules out cases where mature-market demand is sufficiently inelastic that firms prefer to restrict quantities despite the reduction in costs.

[18] Corollary A4 in Appendix A.3 provides conditions guaranteeing this. These are satisfied under constant shape restrictions, e.g. stochastic dominance for log-concave willingness-to-pay distributions (Milgrom and Weber 1982; Chade and Schlee 2012; Chade and Swinkels 2021).

[19] Consumers learn about the generic "product" offered by firms instead of developing brand loyalty.

PROPOSITION 4. *If a monopolist cannot earn positive profits in immature markets, there exists a subgame perfect equilibrium where no firm enters. Moreover, this is the unique extensive-form trembling hand perfect equilibria.*

Importantly, this depends only on immature market primitives and holds even when the mature market is efficient. Static adverse selection in new markets can create missing mature markets regardless of mature market efficiency. The proof is in Appendix B. The no-entry logic extends to dynamic finite-horizon settings, including stochastic transitions from immature to mature states. The key is that firms bearing early losses cannot appropriate future gains from market maturity.

The intuition reveals a new free-riding problem preventing market maturation. Entering an immature market is an investment: the entrant bears negative profits while familiarizing consumers with the product, reducing future adverse selection for all firms.[20] If monopolist profits in the mature market were guaranteed, absorbing these losses might be profitable. But competitors can wait and enter only after this investment is made, competing away the long-run rents that would otherwise cross-subsidize early losses. Consequently, no firm enters.[21]

This logic requires that no positive profits exist in the market's infancy. Unlike settings where low initial demand alone deters entry through scale economies, here the customer pool itself is unprofitable to serve: adverse selection makes the marginal entrant's expected costs exceed any feasible price. To see this, consider the standard profit margin expression from equation (16) (see, also Starc 2014; Kong et al. 2024):

$$p - AC(p) = \frac{1}{\eta_{D,p}} - (AC(p) - MC(p)) \,. \tag{18}$$

Firms may earn negative monopolist profits when the adverse selection wedge, $AC(p) - MC(p)$, exceeds the Lerner markup at all quantities.[22] No price remedies this because higher prices shed the lowest-cost buyers first, leaving average costs weakly higher, so the demand curve lies everywhere below the average cost curve as in the complete unraveling case of Figure 2D.

[20] Indeed, at least one major Canadian auto-insurer offers individuals subsidies (above rating incentives) to enroll in new telematics contracts.

[21] This assumes aggregate state transitions only once contracts exceed a positive threshold, so firms cannot "enter" by charging prices guaranteeing zero profits and no sales.

[22] Kong et al. similarly discuss entry under imperfect competition and derive a threshold condition on the slope of the average cost curve to sustain a given market structure.

3.1. Implications For Market Intervention

If mature markets generate surplus, maturation may be socially optimal despite no-trade in infancy. The competitive allocations can be efficient once markets reach maturity, so phasing out any intervention is eventually optimal. Thus, permanent policy solutions may be inefficient because the primitives evolve over time. Implementing time-varying subsidies, however, requires information that is unavailable when markets are missing.

A simple decentralized alternative is removing barriers to patenting or increasing patent protection length. This raises monopolist profits in immature markets by protecting against free-riders, offering an effective time-varying subsidy that depends only on patent length.[23] Alternatively, increasing regulatory burden on new products—through extensive disclosure requirements or costly reporting mandates—can slow diffusion across firms (Benhabib et al. 2021), lengthening the period of market power.[24]

Examples of Phased Intervention. This logic finds empirical analogues in several insurance markets. The ACA's temporary reinsurance program (2014–2016) and risk corridors transferred funds to plans with high-cost enrolees during the exchanges' early years, absorbing adverse selection while the risk pool stabilized; both were legislated with sunset provisions and phased out on schedule. The Dutch basic health insurance reform followed a similar arc: ex post cost compensation for insurers was used to prevent adverse selection from destabilizing the market at launch, then gradually replaced by a permanent ex-ante risk-equalization system as the necessary data infrastructure matured.[25] The Terrorism Risk Insurance Act (2002) provides another instance: a government backstop was introduced when private terrorism reinsurance markets collapsed, and the government's cost-share has been progressively reduced over successive reauthorizations. In each case, the temporary intervention addressed the selection or uncertainty barrier to entry that the permanent market structure could not yet resolve. This is precisely the role the model assigns to time-varying policy.

[23] See Williams (2017) for a survey of other considerations in optimal patent design. Patents additionally require information disclosure, which can help determine optimal patent length.

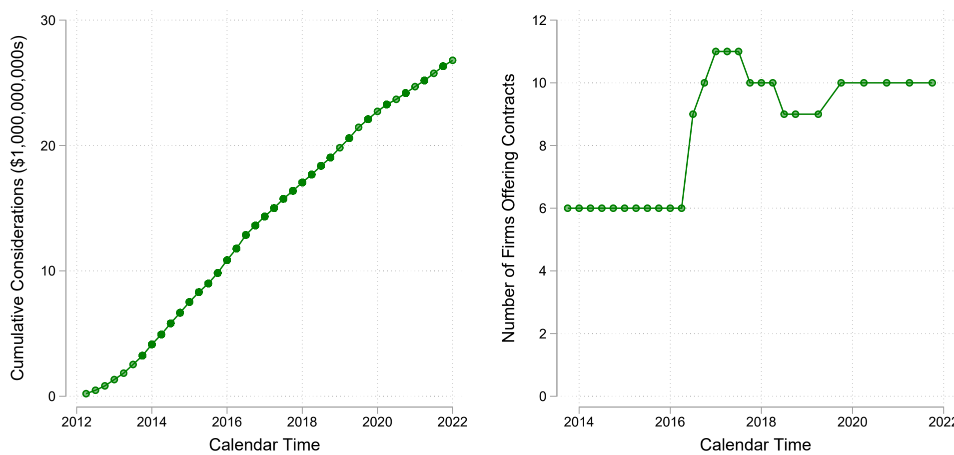
[24] Empirical evidence highlights regulation's role in suppressing innovation (Griffith and Macartney 2014; Aghion et al. 2023).

[25] See van de Ven and Schut (2011) for an overview of the Dutch risk-equalization system and McGuire et al. (2020) for a comparative discussion.

4. Identifying Adverse Selection Among Early Adopters: Evidence from the Market for Deferred Income Annuities

I now provide evidence that early adopters are more adversely selected. As there is no data on missing markets, I instead study the comparison between different stages of market maturity and show how they align with testable implications outlined above. Specifically, I focus on deferred income annuities – a product similar to a traditional annuity, but where payouts do not begin until a date (or age) agreed in the future. Hence, they provide deferred income.

Data on the aggregate size of the market, as measured by the dollar value of new annuity considerations, is collected from the quarterly reports of the Life Insurance Marketing and Research Association.[26] Figure 4A shows how cumulative sales evolved over time. At the start of this period the market for deferred income annuities was small, totaling around $200 million and accounting for less than 1% of the overall fixed income annuity market (Panel B). The market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size.



(A) Cumulative Deferred Income Annuities Considerations

(B) Distinct Firms Operating in the Market for Deferred Income Annuities

FIGURE 4. Growth In The Deferred Income Annuities Market

To study the equilibrium relation between price, quantity, and selection I complement this data on aggregate quantities with hand-collected quotes for individual

[26] Specifically, the U.S. Individual Annuity Sales Survey published by the LIMRA Secure Retirement Institute. Sales are not reported pre-2012 as the size of the deferred income annuities market is minimal.

annuity policies sourced from *Annuity Shopper* magazine between 2013 and 2021.[27] The Annuity Shopper magazine was a longstanding product in the annuity and life insurance industry, targeting potential annuitants as well as financial advisors and Human Resources professionals. Consequently, it is in the interests of life insurers to provide quotes on their products.[28]

In each period, national insurance companies provide quotes for a slate of policies. I collect a detailed quarterly panel dataset of the annuity payouts offered by insurers for homogeneous contract terms with 5,882 observations. These policies vary in the payouts they offer, the life insurer who provides the annuity, the age at which the annuity begins to provide an income stream, as well as the purchaser's current age and gender. In each period there are up to 28 different quotes available for each insurer varying in their contract terms and I focus on single life non-refundable policies without a guarantee period (Einav et al. 2010, study adverse selection in the guarantee decision). An important advantage of this empirical setting is that the product is homogeneous. This differs from health insurance, a common empirical setting, where differences in in- and out-of-network providers lead to differentiated products.

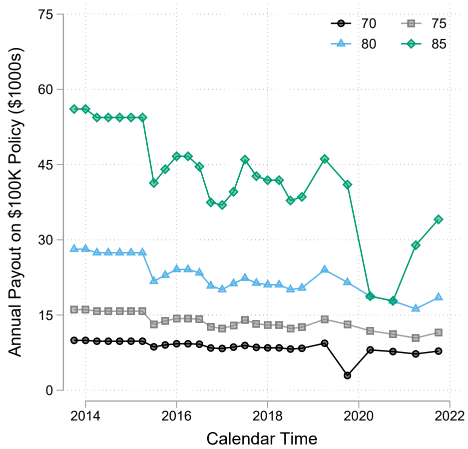
Insurers price annuities for men and women separately, reflecting differences in their survival risk. In addition, annuities are priced separately by the age at which the consumer is purchasing the annuity contract and the deferral age at which the contract begins paying out. Quotes are available for three purchase ages (45, 55 and 65) as well as 5 different deferral ages (65, 70, 75, 80, and 85). Given policy characteristics at a point in time, quotes provide the annual payout offered per $100,000 paid for the deferred income annuity.

Figure 4B shows how the number of firms offering deferred income annuities changes over time. Broadly, there are two regimes: first, a period where the number of entrants is stable at six distinct insurance firms; and second, a period averaging 10 firms. This expansion in the number of operating firms coincides with reforms to the fiduciary duty of brokers operating in brokerage markets (Egan et al. 2022).

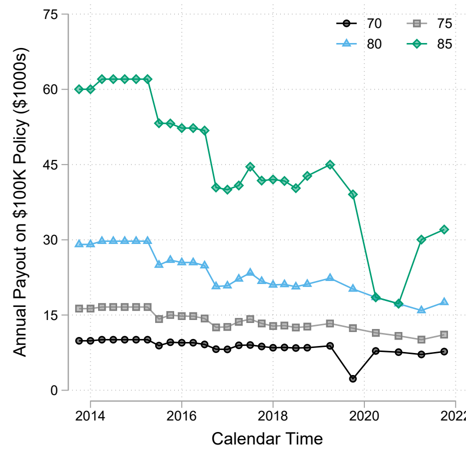
Figure 5 shows payouts from annuities purchased at the age of 65 with policies offered by American General shown in the left column and policies offered by New York Life shown in the right column, two of the largest national insurers. The first row shows policies for men and the second row shows policies for women with each line

[27] Previously The Annuity Shopper's Buyers Guide. Before the third quarter of 2013 quotes for deferred income annuities are not available and, sadly for retirement researchers, it ceased to operate in 2021.

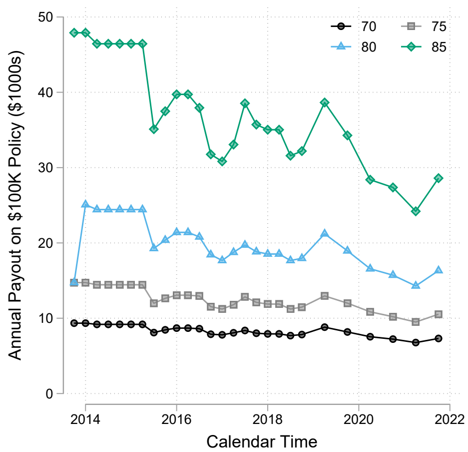
[28] Indeed, over 40 large US insurers provided quotes for a variety of products in the July 2015 issue. Appendix C provides an example testimonial.



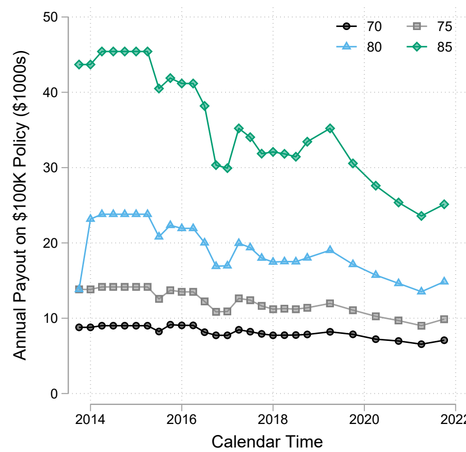
(A) American General (Men)



(B) New York Life (Men)



(C) American General (Women)



(D) New York Life (Women)

FIGURE 5. Average Deferred Income Annuity Payouts at Purchase Age 65 By Insurer

corresponding to a deferral age from age 70 to 85. Three important facts emerge from this comparison. First, there are clear differences in policy generosity by gender. Second, there are systematic differences by the age at which the contract begins paying out. Policies with later deferral dates, which expose consumers to greater mortality risk, offer larger payouts on average and systematic differences across men and women are consistent with differences in their expected longevity. Finally, there are small differences across insurers. Appendix C repeats this exercise conditioning on deferral age, with similar patterns. Common market-level trend in payouts reflects the opportunity cost of foregone investment (Mitchell et al. 1999) and a changing annuitant pool which highlights the potential for selection to drive pricing decisions.

I now turn to measuring selection and maturity, the key quantities of interest, before describing the empirical strategy used to test the implications of the model in Section 2.

4.1. Measuring Selection and Market Maturity

Each policy offered to annuitants at a purchase age, a, of gender, g, is indexed by the deferral age, d, at which payouts begin, the identity of the insurance firm, i, as well as the date, t, at which it is purchased. Each annuity, $\{a, d, g, i, t\}$, offers a fixed income A after the deferral age. I follow Finkelstein and Poterba (2002) and adopt a money's worth calculation to infer selection from annuity payouts. To do this, I calculate the expected present discounted value of the associated annuity income stream

$$EPDV a, d, g, i, t = \frac{1}{\chi_a(g, t-a)} \sum_{h=a}^{110} \frac{A \times \mathbb{1}[h \geq d] \times \chi_h(g, t-a)}{\prod_{j=1}^{h-a}(1 + i_j(t))}, \tag{19}$$

where $\chi_h(g, t-a)$ is probability of surviving until age h as a function of annuitant gender and birth cohort $t-a$. The insurer's interest rate over a horizon j from date t is $i_j(t)$. From this I compute the money's worth value of the annuity (its implicit price):

$$MW_{a,d,g,i,t} = \frac{1}{100,000} EPDV_{a,d,g,i,t}. \tag{20}$$

A money's worth value less than one indicates that the annuity is expected to pay out less than the upfront cost and vice-versa. In the absence of annuity product-specific mortality tables (see Finkelstein and Poterba 2004 for evidence of differential mortality across products) the money's worth value can be used to infer deviations from the mortality table used to calculate it. In this sense, it allows researchers to infer differential mortality and the presence of adverse selection (see also Warshawsky 1988; Mitchell et al. 1999; Poterba and Solomon 2021).[29] I build a measure of selection using the per dollar deviation from the expected present discounted value under zero adverse selection using population mortality tables.

In the baseline construction of money's worth I use the BBB corporate bond yield at 10-year maturity to determine the insurer's interest rate, i_j, and cohort life tables for the age a population to determine mortality risk $\chi_a(g, t-a)$ at a given point in time (linearly interpolating between successive cohort life tables). The deviation from actuarially

[29] Cannon and Tonks (2016) argue that prudent insurers may deviate from actuarially fair pricing when future mortality risk is uncertain. The empirical strategy below allows for this.

fair pricing provides a measure of implicit price and implied adverse selection for each contract. Using gender-specific mortality to calculate money's worth, the implied adverse selection is given by

$$S_{a,d,g,i,t} = 1 - MW_{a,d,g,i,t}, \tag{21}$$

where a larger positive value indicates greater adverse selection.

The testable implications of Section 2 relate adverse selection and market maturity. I proxy unobserved market maturity using the natural logarithm of cumulative sales in the market for deferred income annuities up to that point in time (Figure 4A), $\ln Q_t^{DIA}$. As the measure of new considerations used to construct cumulative sales is not available at a finer disaggregation I use total sales for all deferred income annuities, rather than sales at the characteristic level. This has an intuitive interpretation as a measure of broad market popularity that corresponds to maturity discussed in Section 2. Using a log transformation implies that there is a concave-increasing relationship between maturity and cumulative sales of deferred income annuities and allows for an elasticity interpretation of the relationship between adverse selection and market maturity.

4.2. Empirical Strategy

Given this measure of adverse selection for each contract and a suitable proxy for market maturity, I now describe the empirical strategy used to test the model predictions. A key threat to identifying the effect on adverse selection is that alternative strategic considerations (e.g., dynamic price discrimination) in maturing markets may generate observationally equivalent pricing patterns even absent adverse selection. Thus, the effect of maturity on pricing may not capture the effect of adverse selection.

Absent private information, it would not be possible for adverse selection to change as the market evolves. Therefore, to address these identification concerns and directly test the model's implications, I use differences in available private information by age to exploit variation in the potential for adverse selection. Specifically, as health inequality increases by purchase age (see, e.g., Hosseini et al. 2022; Russo et al. 2024), private information about the future mortality of potential annuitants is also increasing. Equation 10 above formalizes this simple intuition which extends to the general model.

This motivates an empirical strategy that exploits differences in the effects of market maturity by purchase age. I leverage the differential effect of market maturity between those age 45 and those aged 55 and 65. The key identifying assumption is a form of parallel trends assumption: (conditional on other controls) strategic considerations, other pricing decisions, administrative costs, profit rates, and other price-relevant factors would evolve similarly with market maturity if 55- and 65-year-olds had the same level of mortality-relevant private information as 45-year-old consumers. This strategy is implemented by estimating the following specification

$$\begin{aligned}\ln S_{a,d,g,i,t} = \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda \ln q_t^{FIA} + \beta \ln Q_t^{DIA} \\ + \delta \mathbb{1}[a > 45] \times \ln Q_t^{DIA} + u_{a,d,g,i,t},\end{aligned} \tag{22}$$

where γ_a denotes purchase-age-specific fixed effects, γ_d denotes deferral-age-specific fixed effects and, γ_g denotes gender-specific fixed effects. Collectively, these fixed effects capture systematic differences in selection or administrative costs by contractible plan characteristics that do not vary over time. β captures how pricing evolves with market maturity for age 45 purchasers and δ captures the differential effect for groups with more private information. This is akin to a parametric difference-in-differences estimator as it exploits differences in the within purchase age selection as the market matures.

The parameter of interest is δ which measures the differential effect of market maturity on pricing for groups with more private information and, hence, a greater propensity for adverse selection. A key testable implication of the model outlined above is that adverse selection declines as markets mature. This corresponds to $\delta < 0$.

As the market matures, other aggregate factors may simultaneously drive differences in the choices of annuitants or the composition of the pool of potential annuitants including population-level mortality improvements, differences in the financial resources of retirees over time, or the effect of legislative reform (such as the change to broker fiduciary duty studied by Egan et al. 2022). These factors may be correlated with a positive time trend in market maturity and bias the estimate of maturity on adverse selection. To minimize this threat to identification, I control for contemporaneous demand in other fixed income annuity products excluding deferred income annuity considerations (Figure A2, Appendix C), $\ln q_t^{FIA}$, which absorbs time-varying shocks affecting all fixed income annuity products.

Similarly, changes in average prices over time may reflect changes in the composition of insurers due to entry and exit. If insurers serve different populations due to heterogeneous geographic penetration or face different administrative costs, this would confound the effect of market maturity. I control for insurer-specific fixed effects which account for compositional changes in firms.[30]

In addition to the specification in equation (22), I also estimate the following:

$$
\begin{aligned}
\ln S_{a,d,g,i,t} = & \gamma_a + \gamma_d + \gamma_i + \gamma_g + \lambda \ln q_t^{FIA} + \beta \ln Q_t^{DIA} \\
& + \delta_{55} \mathbb{1}[a = 55] \times \ln Q_t^{DIA} + \delta_{65} \mathbb{1}[a = 65] \times \ln Q_t^{DIA} + u_{a,d,g,i,t}.
\end{aligned} \tag{23}
$$

This differs in explicitly examining the heterogeneity between consumers who purchase annuities at age 55 and those who purchase at age 65. Examining this heterogeneity allows for a direct empirical test of a second testable implication, that markets with greater (cost-relevant) private information experience larger reductions in adverse selection as markets mature ($\delta_{65} < \delta_{55} < 0$).

Under the identifying assumption, non-selection pricing factors evolve similarly across purchase ages conditional on controls, so β absorbs these common dynamics for the low-private-information group. The differential parameters δ then isolates the selection channel, which operates only through private information. As the market matures, this channel can generate a differential decline across age groups consistent with diminishing adverse selection. When private information is less important for entry and pricing, the effect is necessarily smaller for groups possessing more of it. Both the monotonic pattern in Table 2 and direct tests of private information corroborate this, confirming that the differential scales with the intensity of the mechanism.

4.3. Adverse Selection Among Early Adopters

Table 1 reports the results from estimating equation (22). The first column only controls for fixed differences across purchase ages. The estimated effect of additional private information on the elasticity of selection (as a deviation from actuarially fair pricing) with respect to market maturity is negative. The baseline estimate for age 45 purchasers, $\hat{\beta}$, is positive, reflecting secular pricing trends not fully absorbed by the aggregate Fixed Income Annuity control: precisely why the design nets out those common trends. Under parallel trends, these common dynamics cancel, and δ isolates the selection component. A 1% increase in past cumulative sales of deferred income annuity products is associated with a –0.177 percentage reduction in selection over and above strategic price dynamics for purchase ages with less private information. Thus, as markets mature, adverse

[30]Although annuities are homogeneous, insurers may differ in other ways such as customer service or integration with other financial services. The insurer fixed effect also controls for time-invariant product differentiation. Section 2.5 extends the theoretical predictions to this setting.

	(1)	(2)	(3)	(4)	(5)
$\ln Q_t$	0.411^{***}	0.445^{***}	0.270^{***}	0.465^{***}	0.287^{***}
	(0.034)	(0.031)	(0.029)	(0.031)	(0.030)
$\mathbb{1}[a > 45] \ln Q_t$	-0.177^{***}	-0.214^{***}	-0.220^{***}	-0.227^{***}	-0.233^{***}
	(0.037)	(0.033)	(0.031)	(0.033)	(0.031)
FEs					
Gender	*No*	*Yes*	*Yes*	*Yes*	*Yes*
Deferral Age	*No*	*Yes*	*Yes*	*Yes*	*Yes*
Insurer	*No*	*No*	*No*	*Yes*	*Yes*
FIA Demand	*No*	*No*	*Yes*	*No*	*Yes*
N	5,831	5,831	5,831	5,831	5,831
R^2	0.137	0.390	0.459	0.416	0.484

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ in equation (22). Implied selection is recovered using equation (21) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All specifications additionally control for purchase age indicators.

TABLE 1. Adverse Selection and Market Maturity

selection decreases. This is evidence of larger adverse selection among early adopters.

The estimated effect is economically significant and statistically significant at the 1% level; rejecting the null hypothesis of no or advantageous selection. Additionally, I also find that measured selection is on average larger for older purchase ages with more private information.

The second column adds additional controls for important contract characteristics: gender and deferral age. As the pricing of annuities varies across these characteristics there is a large increase in the overall explanatory power as measured by the R-squared. However, there is only a small impact on the estimated decline in adverse selection. The elasticity increases, but is overall similar to the previous point estimate. This is because these plan characteristics deliver systematically different pools of annuitants, but evolve similarly over time.[31] The third column adds observed demand for all other fixed income annuities, $\ln q_t^{FIA}$, a proxy for aggregate factors that drive annuity demand and pricing in general. Although this absorbs some of the common effect of market maturity

[31]Estimated effects are qualitatively similar when estimated separately for each gender and deferral age grouping exploiting the same variation in private information across purchase ages. In some cases, these results have less precision (as each subsample is approximately one-tenth of the pooled sample).

on pricing, it does not affect the point estimate of the differential effect because the empirical strategy exploits differences across purchase ages to identify the component driven by selection.

The fourth column instead includes insurer-specific fixed effects to control for changes in the composition of insurers who may face different pools of annuitants or administrative costs. Overall, the point estimate increases slightly although the differences across columns two and four are statistically insignificant. This reflects the fact that these are national insurers who cover the whole US population and, thus, face similar pools of potential annuitants. Although there are differences in administrative costs across insurers (Mitchell et al. 1999; Koijen and Yogo 2015), these are, on average, only weakly correlated with entry into this market. The final column combines the controls in columns three and four. Across all sets of controls, the estimated elasticity is stable in magnitude and statistically significant at the 1% level. The empirical evidence in this market highlights that adverse selection can be higher among early adopters.

	(1)	(2)	(3)	(4)	(5)
$\ln Q_t$	0.411***	0.445***	0.270***	0.465***	0.287***
	(0.034)	(0.031)	(0.029)	(0.031)	(0.030)
$\mathbb{1}[a = 55] \ln Q_t$	−0.091**	−0.127***	−0.132***	−0.139***	−0.144***
	(0.041)	(0.036)	(0.034)	(0.035)	(0.033)
$\mathbb{1}[a = 65] \ln Q_t$	−0.287***	−0.326***	−0.332***	−0.339***	−0.346***
	(0.038)	(0.033)	(0.031)	(0.032)	(0.031)
FEs					
Gender	*No*	*Yes*	*Yes*	*Yes*	*Yes*
Deferral Age	*No*	*Yes*	*Yes*	*Yes*	*Yes*
Insurer	*No*	*No*	*No*	*Yes*	*Yes*
FIA Demand	*No*	*No*	*Yes*	*No*	*Yes*
N	5, 831	5, 831	5, 831	5, 831	5, 831
R^2	0.144	0.397	0.466	0.423	0.491

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient $\delta_5 5$ and $\delta_6 5$ in equation (23). Implied selection is recovered using equation (21) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All specifications additionally control for purchase age indicators.

TABLE 2. Adverse Selection and Market Maturity by Private Information

Table 2 reports the results of estimating the specification in equation (23). The key difference is that this disaggregates the elasticity into separate estimates for age 55 and 65 purchasers. As with the results in Table 1, the parameter estimates differ slightly when controls for contract characteristics are added (columns one compared with two). Again, the point estimates are largely stable across alternative controls.

The second row reports the differential elasticity of selection for age 55 purchasers. These estimates vary between –0.091 and –0.144 and are consistently negative. The third row reports the elasticity of selection for age 65 purchasers. These estimates are larger in magnitude as the pooled effect is the weighted average, varying between –0.287 and –0.346, and are also consistently negative. Like the pooled elasticity they are always statistically significant at the 1% level and reject the null hypothesis of no or advantageous selection in each age group.

The differential decline scales monotonically with purchase age and therefore with private information, and the difference between δ_{65} and δ_{55} is statistically significant. This jointly tests both model predictions and corroborates the identifying assumption of less private information for younger age groups.

4.4. Threats to Identification and Alternative Mechanisms

I assess the key identifying assumption that mortality-relevant private information is larger at older purchase ages. This assumption is crucial to separately identify the dynamics of adverse selection in a maturing market from other time-varying factors.

To do so, I use subjective mortality expectations for a representative sample of potential annuitants, age 50 to 70, from the Health and Retirement Study (see Appendix D for details). Respondents are asked: "What is the percent chance that you will live to be [X] or more?" where the target age X depends on the respondent's current age (75 for those aged 65 or younger, 85 for those aged 70–74, etc.).

Following Hurd and McGarry (2002), I compare realized mortality across individuals with different subjective survival beliefs at different ages. I construct an indicator for 10-year mortality, $M_{i,t+10}$, and estimate the following linear probability model[32]

$$M_{i,t+10} = \beta_0 + \beta_{surv}\text{Belief}_{i,t} + \beta_{Age}\text{Age}_{i,t} + \delta\text{Belief}_{i,t} \times \text{Age}_{i,t} + \beta X_{i,t} + u_{i,t}, \tag{24}$$

where the coefficient δ measures how predictive accuracy varies by age. If older individuals possess more precise private information about their mortality, their stated

[32] I focus on a subsample for whom mortality outcomes are resolved: either dying or being observed at least 10 years later.

beliefs should better predict realized mortality and $\delta < 0$.

	(1)	(2)	(3)	(4)	(5)
$\text{Belief}_{i,t}$	-0.106^{***}	-0.014	-0.012	-0.012	-0.011
	(0.011)	(0.011)	(0.011)	(0.011)	(0.010)
$\text{Belief}_{i,t} \times \text{Age}_{i,t}$	-0.006^{***}	-0.004^{***}	-0.005^{***}	-0.005^{***}	-0.003^{***}
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Controls					
Health	*No*	*Yes*	*Yes*	*Yes*	*Yes*
Demographics	*No*	*No*	*Yes*	*Yes*	*Yes*
Education	*No*	*No*	*No*	*Yes*	*Yes*
Wave FE	*No*	*No*	*No*	*No*	*Yes*
N	109, 368	109, 343	109, 316	109, 305	109, 305
R^2	0.136	0.195	0.204	0.204	0.299

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by individual. Dependent variable is an indicator for death within 10 years. Sample restricted to respondents aged 50–70 for whom mortality outcomes are resolved. Data from the Health and Retirement Study, waves 1992–2018 estimating equation 24. Column 1 includes no controls. Column 2 adds self-reported health indicators. Column 3 adds gender and race. Column 4 adds education. Column 5 adds wave fixed effects, controlling for systematic mortality improvements.

TABLE 3. Mortality-Relevant Private Information and Age

Table 3 reports estimates of δ with age centered at 50 so that β_{surv} captures the effect of survival beilefs at age 50. Once health and demographic controls are included (columns 2–5), the predictive effect at age 50 becomes statistically insignificant as health observables absorb mortality-relevant information for younger respondents. However, the interaction δ remains negative and highly significant across all specifications: beliefs become increasingly predictive with age even conditional on observables. At age 65, the implied marginal effect is approximately -0.06 so that a 10 percentage point higher survival belief predicts a 0.6 percentage point lower 10-year mortality. Since annuity pricing conditions only on age and gender, the full predictive content of beliefs—not just the residual after controls—represents private information from the insurer's perspective. That this predictive power increases with age validates the key identifying assumption.

Response Quality. A potential concern is that older respondents provide less noisy answers due to mortality salience rather than possessing more private information. Average subjective mortality exhibits a "flatness" bias (e.g., Hamermesh 1985; Elder 2013; Foltyn and Olsson 2024) due to focal-point heaping. However, Table A5 (Appendix D) shows that these results are robust—and indeed slightly stronger—when excluding focal responses (0%, 50%, 100%). Moreover, focal response rates do not vary systematically with age in the sample.

4.4.1. Robustness and Alternative Mechanisms

In Appendix C I show that results are not driven by the specific choice of mortality table or the baseline opportunity cost of capital, including when Treasury yields replace the corporate bond rate as the Department for Labor requires for defined contribution plan sponsors. Discounting each horizon at the corresponding risk-free zero-coupon rate mechanically compresses cross-contract variation in money's worth and attenuates the estimates to zero; I discuss this further in Appendix C. The direct evidence from subjective mortality beliefs (Table 3) provides evidence of age-varying private information that does not depend on discount rate assumptions, lending direct support to a selection interpretation of the market-level results.

Differential Markups and Alternative Mechanisms. A concern with interpreting these results as evidence of declining adverse selection is that insurers may charge differential markups across purchase ages that evolve over time. The identification strategy addresses this directly. By exploiting differences in how pricing evolves across age groups, common markup dynamics are absorbed. For the results to be driven by markups rather than selection, markup dynamics would need to differ systematically by purchase age in a way that correlates with private information. Markup compression is plausibly larger for younger, more elastic purchasers, thus the relative price reductions for older purchase ages work in a direction opposite to what entry patterns suggest. Figure 4B shows increasing entry over time, which should compress markups at all ages through competition; yet relative price reductions are largest for older purchasers. Indeed, replacing aggregate demand controls with unrestricted time fixed effects non-parametrically absorbs all time-varying common factors, including any markup dynamics induced by firm entry, leaving the differential estimate across purchase ages virtually unchanged (Appendix C).

Two specific mechanisms could still generate differential dynamics. First, in a new market firms may be learning about the distribution of risks (i.e., cost types) and may do so differentially among the pools of potential annuitants at different purchase ages. This could lead to initial miss-pricing and induce a negative correlation. However, although deferred income annuities are a new product, insurers are experienced in pricing annuity and life-insurance contracts based on the distribution of underlying mortality risk. As equation (21) shows, the actuarial formula for a deferred income annuity payout is almost identical to the formulas for these existing products. They differ only in the timing at which payout begins. Therefore, it would have to be the case that insurers were learning about the declining adverse selection by age as the market matures which is exactly the fundamental prediction of the model.

Second, firms may initially 'invest' in market power, e.g. by undercutting or offering 'sweetheart' deals, and subsequently raise markups to 'harvest' downward sloping residual demand.[33] This would produce a negative contemporaneous correlation between equilibrium prices and demand, a distinct implication to the time path considered here. Additionally, this mechanism has sharp predictions about the declining number of insurers in the market which is at odds with Figure 4. The number of insurers offering these contracts grows over time—inconsistent with price and quantity dynamics driven by this form of strategic behavior.

5. Discussion

This paper identifies a novel dynamic free-rider problem in adversely selected markets. When early adopters are more adversely selected, initial entrants bear losses that later competitors avoid by entering only after the market matures. Because long-run rents are competed away before they can cross-subsidize early losses, entry unravels even when the mature market would be efficient — generating permanently missing markets as a dynamic counterpart to the classic static unraveling of Akerlof (1970).

I microfound these entry incentives in a standard insurance model with multidimensional heterogeneity, providing non-parametric sufficient conditions—stated in terms of copulas governing the joint distribution of costs and willingness-to-pay—for ordering adversely selected markets by equilibrium prices, quantities, selection, and welfare. These conditions accommodate commonly used empirical specifications and map directly into the graphical framework of Einav and Finkelstein (2011), extending its comparative statics to dynamic and cross-market settings. Together they jointly rationalize low utilization, sluggish take-up, and market inactivity despite substantial unmet demand.

Evidence from the deferred income annuity market confirms the mechanism is

[33]This may arise if consumers do not switch products due to search costs. This is unlikely to be of relevance here where the annuitization decision is typically once and for all.

empirically relevant. I exploit age-based differences in mortality-relevant private information, validated using subjective mortality beliefs, to separate the selection component of price changes from secular pricing trends and alternative markup dynamics. This selection component is negative, scales with private information, and declines as the market matures. This is consistent with the model's joint predictions. Although the analysis focuses on insurance, the mechanism applies to any selection market where buyers impose heterogeneous costs on sellers and familiarity with the product grows over time. More broadly, the framework delivers predictions wherever risk premia shift across markets or over time: as Blundell et al. (2026) show in durable goods markets, such shifts compress the lemons penalty, and the comparative statics developed here provide the microfoundations for these related results.

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Online Appendix

Appendix A. Proofs for Section 2

This section provides general versions of the results established under normality in Section 2.3 and extends them to non-parametric families of joint distributions.

A.1. General Proofs and Additional Results for Sections 2.2–2.5: Within Markets

The Role of the MLRP assumption. The MLRP property is widely invoked in studies of incomplete information and has a convenient intuitive interpretation. The proof of Proposition 3 establishes that monotonicity of the cost curves are each available under milder assumptions on the copula that are directly implied by the stronger MLRP.[1]

PROOF OF PROPOSITION 3 . Preliminaries. Let copula C denote the dependence between cost type, π_i, and willingness-to-pay, ω_i. Then the monotone likelihood ratio dependence implies (see, e.g., Nelsen 2006 Theorem 5.2.19):

i. [Component-wise Concavity] $C(u, v)$ is concave in v for all $u \in (0, 1)$.

ii. [Corner Set Monotonicity] $(u-C(u,1-v))/v$ is increasing in v for all $u \in (0, 1)$.

Proof of Part (i). Beginning with the monotonicity of marginal costs – the CDF of the conditional random variable $\pi|\omega = p$ is

$$
\begin{aligned}
F_{\pi|\omega}(\pi|\omega) \equiv \int_{-\infty}^{\pi} f_{\pi|\omega}(u|\omega)du &= \int_{-\infty}^{\pi} \frac{f_{\pi,\omega}(u,\omega)}{f_{\omega}(\omega)}du \\
&= C_2(F_\pi(\pi), F_\omega(\omega)) = g(F_\pi(\pi)),
\end{aligned} \tag{A1}
$$

for a given copula C where $C_2(x, y) = \partial C(x,y)/\partial y$.

As the marginal cost curve is a conditional expectation $MC(p) = E(\pi|\omega = p)$, stochastic dominance of the conditional distributions is a sufficient condition to order marginal costs in price

$$F_{\pi|\omega}(\pi|\omega = v_1) \succeq F_{\pi|\omega}(\pi|\omega = v_2) \;\; \forall v_1 \geq v_2 \tag{A2}$$

$$\longrightarrow MC(v_1) = E(\pi|\omega = v_1) \geq E(\pi|\omega = v_2) = MC(v_2) \;\; \forall v_1 \geq v_2 \, . \tag{A3}$$

[1] A weaker necessary condition for the second property is a log-concave survival function – a common assumption when studying incomplete information (for instance Milgrom and Weber 1982, Chade and Schlee 2012 and Chade and Swinkels 2021).

Substituting equation (A1) gives the following necessary and sufficient condition:

$$C_2(u, F_\omega(v_1)) \leq C_2(u, F_\omega(v_2)) \; \forall v_1 \geq v_2 \textit{ and } \forall u \tag{A4}$$

$$\longleftrightarrow C_2(u, a) \leq C_2(u, b) \; \forall a \geq b \textit{ and } \; \forall u \tag{A5}$$

as $v_1 \geq v_2 \longrightarrow F_\omega(v_1) \geq F_\omega(v_2)$. This is satisfied for all copulas where $C_2(\cdot, \cdot)$ is decreasing in its second argument. Condition (i) of Proposition (3) guarantees C is concave and satisfies this property.

This establishes that $MC(p)$ is increasing in p.

Proof of Parts (ii) and (iii). By the law of iterated expectations, the average cost curve admits the representation

$$AC(p) = E(\pi | \omega \geq p) = E(MC(\omega) \,|\, \omega \geq p)\,. \tag{A6}$$

Since $MC(\cdot)$ is weakly increasing (Part (i)), raising the truncation threshold from v_2 to $v_1 \geq v_2$ shifts the conditional distribution of $\omega | \omega \geq p$ in the sense of first-order stochastic dominance, so

$$AC(v_1) = E(MC(\omega) \,|\, \omega \geq v_1) \geq E(MC(\omega) \,|\, \omega \geq v_2) = AC(v_2) \; \forall v_1 \geq v_2, \tag{A7}$$

establishing Part (ii). Moreover, on the event $\{\omega \geq p\}$, monotonicity implies $MC(\omega) \geq MC(p)$ pointwise, so

$$AC(p) = E(MC(\omega) \,|\, \omega \geq p) \geq MC(p), \tag{A8}$$

establishing Part (iii). □

Corollary A1 (Corner-Set Copula Condition for Average Costs). *The monotonicity of $AC(p)$ in Part (ii) of Proposition 3 can alternatively be established directly from a stochastic dominance ordering. The CDF of the conditional random variable $\pi | \omega > p$ is*

$$F_{\pi|\omega > p}(\pi | \omega > p) \equiv \frac{u - C(u, F_\omega(p))}{1 - F_\omega(p)}. \tag{A9}$$

Substituting equation (A9), this holds whenever the copula satisfies

$$\frac{u - C(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - C(u, F_\omega(v_2))}{1 - F_\omega(v_2)} \; \forall v_1 \geq v_2 \textit{ and } \forall u, \tag{A10}$$

which is condition (ii) of Proposition 3. This copula restriction is the relevant sufficient *primitive for within-market comparisons of average costs in Proposition A3.*

A.2. Additional Results and Examples of Copula

This section shows how to derive the distribution of willingness-to-pay directly from assumptions on cost and risk premia in the linear case. This will depend on the dependence between cost and risk premia, as encoded by the copula $\widetilde{C}$, and marginal distributions. The following results are derived in Cherubini et al. (2011) who also provide extensions to multivariate settings (see also Navarro and Sarabia 2022). These multivariate extensions can also be used to derive expressions for the joint-density of net premia from the underlying densities of risk premia r and choice frictions ε. This extends general results on the convolution of probability distributions.

The distribution function for the ω is given by F_ω

$$F_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c)\partial_1\widetilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c))\,dc \tag{A11}$$

provided that $\lim_{v\to 0^+} \partial_1\widetilde{C}(u,v) = 0$ for all $u \in (0,1)$. This is a mixture over the conditional distribution of $(\tilde{r} + c|\pi = c)$, which is given by[2]

$$F_{\omega|\pi}(\omega|\pi = c) = \partial_1\widetilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)). \tag{A12}$$

We can use this to derive the following joint CDF

$$F_{\pi,\omega}(t,v) = \int_{-\infty}^{t} F_{\omega|\pi}(v|s)\,dF_\pi(s) = \int_{-\infty}^{t} \partial_1\widetilde{C}(F_\pi(s), F_{\tilde{r}}(v - s))\,dF_\pi(s), \tag{A13}$$

which in turn can be used recover C, the copula specifying the dependence between ω and π, by application of Sklar's Theorem:

$$C(u,v) = \int_{-\infty}^{v} \partial_1\widetilde{C}(x, F_{\tilde{r}}(F_\omega^{-1}(u) - F_\pi^{-1}(x)))\,dx. \tag{A14}$$

There are not closed form results for this integral as it depends on the marginal densities and the copula. However, we can additionally derive the following probability density functions which will prove useful in evaluating its properties:

$$f_{\omega|\pi}(\omega|\pi = c) = f_{\tilde{r}}(\omega - c)\partial_{1,2}\widetilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c)) \tag{A15}$$

$$f_\omega(\omega) = \int_{-\infty}^{\infty} f_\pi(c) f_{\tilde{r}}(\omega - c)\partial_{1,2}\widetilde{C}(F_\pi(c), F_{\tilde{r}}(\omega - c))\,dc \tag{A16}$$

[2]Theorem 2.3 Navarro and Sarabia 2022 shows that this admits a distortion function representation.

and the following corollary uses these representations to re-express Proposition 3 in terms of the joint distributions of π and $\tilde{r}$.

COROLLARY A2 (MLRP). *The monotone-likelihood ratio property for willingness-to-pay, ω, and cost type, π, is equivalent to the following property for the cost types and risk premia $\tilde{r}$:*

$$\frac{f_{\tilde{r}}(\omega-\pi)\partial_{1,2}\widetilde{C}(F_\pi(\pi),F_{\tilde{r}}(\omega-\pi))}{f_{\tilde{r}}(\omega-\pi')\partial_{1,2}\widetilde{C}(F_\pi(\pi'),F_{\tilde{r}}(\omega-\pi'))} \geq \frac{f_{\tilde{r}}(\omega'-\pi)\partial_{1,2}\widetilde{C}(F_\pi(\pi),F_{\tilde{r}}(\omega'-\pi))}{f_{\tilde{r}}(\omega'-\pi')\partial_{1,2}\widetilde{C}(F_\pi(\pi'),F_{\tilde{r}}(\omega'-\pi'))} \ \forall \pi' \geq \pi \text{ and } \omega' \geq \omega$$

Thus, all joint distributions satisfying this property inherit properties (i) - (iii) in Proposition 3.

COROLLARY A3 (Primitive Sufficient Condition for the Additive Index). *Suppose $\omega = \pi + \tilde{r}$ and let $h_\pi(r) \equiv f_{\tilde{r}|\pi}(r|\pi)$ denote the conditional density of $\tilde{r}$ given π. If for every $\pi' > \pi$:*

i. the family $\tilde{r}|\pi$ satisfies the monotone likelihood ratio property in $(\tilde{r}, \pi)$, i.e.

$$\frac{h_{\pi'}(r)}{h_\pi(r)} \quad \text{is weakly increasing in } r; \tag{A17}$$

ii. for each π, the density $h_\pi(\cdot)$ is log-concave.

Then the conditional density $f_{\omega|\pi}(\omega|\pi)$ satisfies the monotone likelihood ratio property in (ω, π). Hence Proposition 3 holds for ω, and therefore also for any strictly increasing transformation $\omega = g(\omega)$.

PROOF. Write $\Delta = \pi' - \pi > 0$ and $r = \omega - \pi$. Since $f_{\omega|\pi}(\omega|\pi) = h_\pi(\omega - \pi)$, the likelihood ratio can be decomposed as

$$\frac{f_{\omega|\pi}(\omega|\pi+\Delta)}{f_{\omega|\pi}(\omega|\pi)} = \underbrace{\frac{h_{\pi+\Delta}(r-\Delta)}{h_{\pi+\Delta}(r)}}_{\text{increasing in } r \text{ by (ii)}} \cdot \underbrace{\frac{h_{\pi+\Delta}(r)}{h_\pi(r)}}_{\text{increasing in } r \text{ by (i)}} . \tag{A18}$$

The second term is increasing in r by assumption (i). The first term is increasing in r by log-concavity of $h_{\pi+\Delta}$, since log-concavity implies $\log h_{\pi+\Delta}(r-\Delta) - \log h_{\pi+\Delta}(r)$ is increasing in r.[3] Therefore the full likelihood ratio is increasing in ω, establishing MLRP for $\omega|\pi$. The stated conclusions then follow from the MLRP Corollary above. □

This result decouples the restriction from the joint law of (π, ω) to a statement about the conditional family $\tilde{r}|\pi$, which is more directly interpretable in terms of model primitives. Condition (i) requires that higher cost types draw risk premia from

[3]This is a standard result that log-concavity of a density implies monotonicity of its hazard rate; see, e.g., Milgrom 1981.

stochastically higher distributions in the likelihood ratio sense; condition (ii) is a regularity condition on the shape of each conditional density that is satisfied by all log-concave families, including the normal, exponential, uniform, and logistic.

I now turn to discussing examples of common parametric copulas and the conditional distributions of ranks (summarized by their derivatives). While this allows for convenient theoretical analysis, it is worth emphasizing, however, that in empirical settings copulas can be estimated directly from the data using non-parametric tools. This is an inexhaustive list and other copula family also abound.[4,5]

Example 1: Gaussian copula. The Gaussian copula is a family of copulas that vary in their correlation parameter ρ. All multivariate Normal distributions have Gaussian copulas and Normal marginals. However, not all normally distributed marginals have Gaussian copulas. The bivariate copula is defined as:

$$C(u, v; \rho) = \int_{-\infty}^{\Phi^{-1}(u)} \int_{-\infty}^{\Phi^{-1}(v)} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left[-\frac{s^2 - 2\rho st + t^2}{2(1-\rho^2)}\right] ds\, dt \tag{A19}$$

$$= \int_{-\infty}^{\Phi^{-1}(u)} \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{s^2}{2}\right] \Phi\left[\frac{\Phi^{-1}(v) - \rho s}{\sqrt{1-\rho^2}}\right] ds \tag{A20}$$

with first derivative given by (it is symmetric)

$$\frac{d}{du} C(u, v; \rho) = \frac{1}{\sqrt{2\pi}} \exp\left[-\frac{\Phi^{-1}(u)^2}{2}\right] \cdot \frac{d\,\Phi^{-1}(u)}{du} \cdot \Phi\left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}}\right] \tag{A21}$$

$$= \Phi\left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}}\right] \tag{A22}$$

and cross-partial derivative (or copula density)

$$\frac{\partial^2}{\partial u \partial v} C(u, v; \rho) = \frac{1}{\sqrt{1-\rho^2}} \frac{d}{dv} \Phi^{-1}(v) \Phi' \left[\frac{\Phi^{-1}(v) - \rho\Phi^{-1}(u)}{\sqrt{1-\rho^2}}\right] \tag{A23}$$

[4] Moreover, the convolution of copulas above is closed under mixtures which, with sufficient mixing, allows for almost arbitrary dependence using parametric families.

[5] A less common copula in statistical applications, but highly relevant in economic applications is the Burr copula. This is characterized by joint-Burr distributions which exhibit similar jointness properties to the normal distribution. Moreover, univariate Burr distributions have extensive use as a generalized version of the Pareto distribution to model income distributions with long tails (see, for instance, Singh and Maddala 1976).

$$= \frac{1}{\sqrt{1-\rho^2}} \exp\left[\frac{-\rho^2\Phi^{-1}(v)^2 + 2\rho\Phi^{-1}(u)\Phi^{-1}(v) - \rho^2\Phi^{-1}(u)^2}{2\left(1-\rho^2\right)}\right]. \tag{A24}$$

Substituting into Equation A12 gives the following expression for the conditional CDF in this example:

$$F_{\omega|\pi}(\omega|\pi = c) = \Phi\left[\frac{\Phi^{-1}(F_\pi(c)) - \rho\Phi^{-1}(F_{\tilde{r}}(\omega - c))}{\sqrt{1-\rho^2}}\right], \tag{A25}$$

which highlights that the normality of the conditional density is a special property. It requires both the specific Gaussian copula and the normally distributed marginals.

Figure A1 shows how the parametric specification of the copula effects the joint distribution when the marginal distribution are fixed. The left columns shows how the joint distribution of π and $\tilde{r}$ changes with the correlation parameter. In both cases I assume π and $\tilde{r}$ are log normally distributed with their logs following standard normals. Comparing panels A and C shows the role of the correlation parameter for the gaussian copula. As the correlation increases, it stretches the density towards the upper right corner.

Example 2: Gumbel Copula. The Gumbel copula has heavier tails than the normal distribution and is also asymmetric, with more weight in the right tail. It's left tail, however, is similar to the normal distribution. The bivariate copula is defined (for $\delta \geq 1$) as:

$$C(u, v; \delta) = \exp\left(-\left[(-\log u)^\delta + (-\log v)^\delta\right]^{1/\delta}\right) \tag{A26}$$

$$= \exp\left(-A^{1/\delta}\right) \quad \text{where} \quad A = (-\log u)^\delta + (-\log v)^\delta \tag{A27}$$

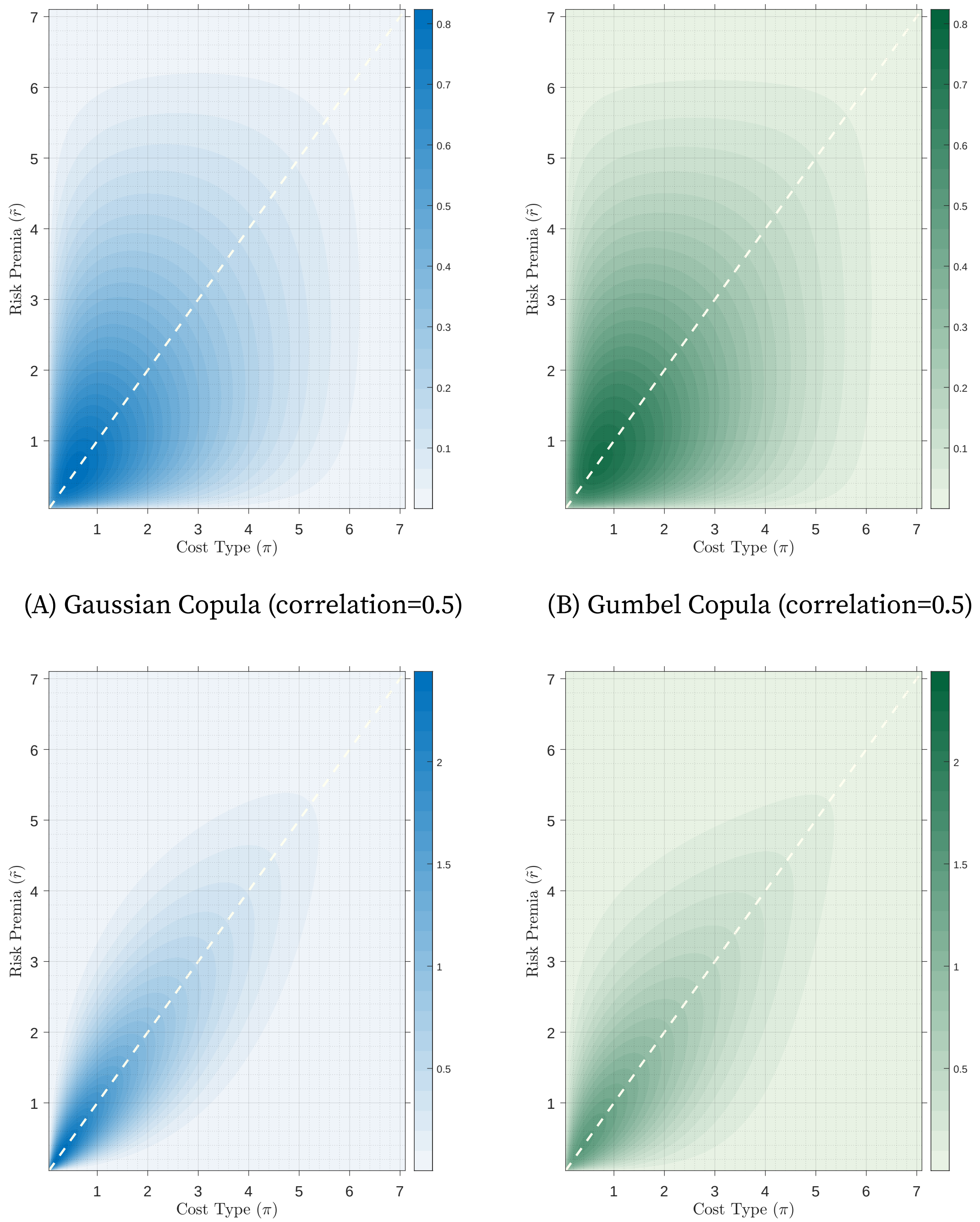
with first derivative given by (it is symmetric)

$$\frac{\partial}{\partial u} C(u, v; \delta) = \exp\left(-A^{1/\delta}\right) \cdot \frac{(1 + (-\log v/-\log u)^\delta)^{1/\delta - 1}}{u} \tag{A28}$$

and, letting $A = (-\log u)^\delta + (-\log v)^\delta$, the cross-partial derivative is

$$\frac{\partial^2}{\partial u\, \partial v} C(u, v; \delta) = C(u, v; \delta)\, \frac{(-\log u)^{\delta-1}\,(-\log v)^{\delta-1}}{u\, v} A^{1/\delta - 2}\left(A^{1/\delta} + \delta - 1\right). \tag{A29}$$

Comparing the right column of Figure A1, which shows the Gumbel copula, to



(A) Gaussian Copula (correlation=0.5)

(B) Gumbel Copula (correlation=0.5)

(C) Gaussian Copula (correlation=0.9)

(D) Gumbel Copula (correlation=0.9)

FIGURE A1. Copula and Joint Densities

Notes: Panels vary the copula (and parameter controlling correlation) for the rank dependency between cost type (π) and risk premia ($\tilde{r}$). In all panels the marginal distributions of cost type (π) and risk premia ($\tilde{r}$) are distributed log-normally with a mean of 0 and unit variance.

the left highlights differences induced in the joint density by choice of the copula. At lower correlations, panels A and B, the two copulas deliver more similar joint-densities. However, panel B, using the Gumbel copula, reveals heavier tails of the joint distribution. This is even more pronounced for higher correlations shown in panels C and D. Here, like the Gaussian copula , panel D stretches the density towards the upper-right corner of the figure. In contrast though, the normal exhibits softer contours. Instead, the surface of the joint density is pulled "tauter" under the Gumbel copula and more mass is concentrated in the joint tail. Thus the risk is more jointly heavily tailed.

Example 3: The Independent Copula. The independent copula describes the case where the two random variables are independent. The relevant formulae are:

$$C(u, v) = uv \tag{A30}$$

$$\frac{\partial}{\partial u} C(u, v) = v \tag{A31}$$

$$\frac{\partial^2}{\partial u \partial v} C(u, v) = 1. \tag{A32}$$

A.3. General Proofs and Additional Results for Sections 2.2–2.5: Across Markets

The Role of the Constant dispersion assumption. Constant dispersion in willingness-to-pay is required for the normal distribution as the support is unbounded. Equivalent results are available with truncated tails (Levy 1982). It is not required for the general results under arbitrary marginals.

A.3.1. Demand Contraction: Additional Results

Demand contractions are immediate from assumptions on willingness-to-pay. Instead, I begin by proving demand contraction under the following alternative assumptions on the sub components of willingness-to-pay:

ASSUMPTION A1. *The perceived risk premia (net of any behavioral influences) of insurance contracts are lower for immature markets.* $\tilde{r}^{MM} \succeq \tilde{r}^{IM}$

ASSUMPTION A2. *Risk types, π_i, have identical marginal distributions in mature and immature markets.*

and

ASSUMPTION A3. *Additionally, assume that resorting satisfies the following conditions for the copulas $\widetilde{C}$, the bivariate copula denoting the dependence between π_i and $\tilde{r}_i$*

i. The partial derivatives of $\widetilde{C}$ satisfy $\frac{\partial \widetilde{C}^{MM}(u, F_{\tilde{r}}^{MM}(\tilde{r}))}{\partial u} \leq \frac{\partial \widetilde{C}^{IM}(u, F_{\tilde{r}}^{IM}(\tilde{r}))}{\partial u} \quad \forall\, u \in [0,1]$ *and* $\tilde{r}$.

PROPOSITION A1 (Demand Contraction). *If $\omega_i = u(\pi_i + \tilde{r}_i)$, willingness-to-pay is on average larger and demand expands in mature markets.* $D^{IM}(p) \leq D^{MM}(p) \quad \forall\, p$

PROOF OF PROPOSITION A1. First consider linear utility with $\omega_i = \pi_i^{IM} + \tilde{r}_i^{IM}$. Assumptions A1 - A3 guarantee that the conditions in Theorem 3.3 in Navarro and Sarabia (2022) are satisfied. A direct application gives

$$\omega_i^{IM} = \pi_i^{IM} + r_i^{IM} + \varepsilon_i^{IM} \preceq \pi_i^{MM} + r_i^{MM} + \varepsilon_i^{MM} = \omega_i^{MM} \longrightarrow F_\omega^{MM}(p) \leq F_\omega^{IM}(p)\ \forall\, p\,. \quad \text{(A33)}$$

As the usual stochastic order is preserved under increasing transformations, the stated proposition then follows immediately for any increasing $u(\cdot)$ □

This level effect does not change the identity of marginal (and inframarginal) consumers at any quantity demanded is the same and the equilibrium price increases and demand falls (See Proposition 1). Similar arguments apply to assumptions on r and ε.

Moreover, when there is *only* a level effect the following lemma shows that the utility function can be generalized to allow for arbitrary substitution patterns between π_i.

LEMMA A1 (Arbitrary Substitutes). *When there is no re-sorting ($C^{IM} = C^{MM}$), Proposition A1 can be extended to any willingness-to-pay satisfying*

$$\omega_i = u(\pi_i, \tilde{r}_i) \tag{A34}$$

where $u(\cdot, \cdot)$ is a function increasing in its arguments.

PROOF OF LEMMA A1. Identical in distribution satisfies $\pi^{MM} \succeq \pi^{IM}$. Thus, each element of the vector $(\pi_i, \tilde{r}_i)$ in mature markets is larger in the usual stochastic order than in immature markets. It follows that the vector $\mathbf{X}^{IM} \preceq \mathbf{X}^{MM}$ in the usual *multivariate* stochastic ordering (Shaked and Shanthikumar 2007, Theorem 6.B.14).

This order is preserved for any increasing function $u : \mathcal{R}^2 \to \mathcal{R}$, then $u(\mathbf{X}^{IM}) \preceq u(\mathbf{X}^{MM})$. Thus, $F^{IM}_{\omega}(p) \geq F^{MM}_{\omega}(p)\ \forall\, p$ and the stated lemma follows. This result extends to any multi-dimensional vector (for example r and ε) under corresponding assumptions on their marginals. □

For the special case of the normal distribution, demand contractions are tightly linked to the price elasticity of consumers in immature and mature markets. I denote this $\tilde{\eta}$ to distinguish from η which denotes the semi-elasticity in the main text. The following corollary formalises this for the general case:

COROLLARY A4 (Price Elasticity of Demand). *When willingness-to-pay is smaller in the hazard rate order in mature markets, that is*

$$\frac{f^{IM}(\omega)}{1 - F^{IM}(\omega)} \geq \frac{f^{MM}(\omega)}{1 - F^{MM}(\omega)}\ \forall\, p, \tag{A35}$$

then demand is more price sensitive in immature markets.

PROOF. This follows from substituting the definition of demand into the usual formula for the elasticity,

$$\tilde{\eta}_{D,p} = D'(p)\frac{p}{D(p)} = -f(p)\frac{p}{1 - F(p)}, \tag{A36}$$

which implies in the normal case

$$\tilde{\eta}^{IM}_{D,p} - \tilde{\eta}^{MM}_{D,p} \propto - \underbrace{\left(\frac{\phi(\frac{p-\mu^{IM}_{\omega}}{\sigma_{\omega}})}{1-\Phi(\frac{p-\mu^{IM}_{\omega}}{\sigma_{\omega}})} - \frac{\phi(\frac{p-\mu^{MM}_{\omega}}{\sigma_{\omega}})}{1-\Phi(\frac{p-\mu^{MM}_{\omega}}{\sigma_{\omega}})} \right)}_{\text{level effect}\geq 0} \leq 0 \tag{A37}$$

The inverse Mills ratio and the difference in the inverse Mills ratio are always positive by the log concavity of the normal distribution ensuring non-negativity. Thus consumers are more price elastic in immature markets. □

A.3.2. Cost Expansion: General Results

Section 2.3 establishes cost expansion under normality. The following results generalise these to arbitrary marginal distributions under the conditions of Section 2.4.

LEMMA A2 (Adverse Selection In Levels). *When there is no re-sorting ($C = C^{IM} = C^{MM}$) marginal costs are higher at any price. However, when re-sorting occurs, effects on marginal cost are ambiguous.*

PROOF OF LEMMA A2. Analogously to the proof of Proposition 3

$$F^{IM}_{\pi|\omega}(\pi|\omega) \succeq F^{MM}_{\pi|\omega}(\pi|\omega) \longrightarrow MC^{IM}(p) \geq MC^{MM}(p)\ \forall p. \tag{A38}$$

As the unconditional first order stochastic dominance of private valuations in mature and immature markets implies $F^{MM}_{\omega}(\omega) \leq F^{IM}_{\omega}(\omega)$, (A38) holds if and only if the copulas satisfy

$$C^{IM}_2(u,a) \leq C^{MM}_2(u,b)\ \forall a \geq b \text{ and } \forall u. \tag{A39}$$

This is trivially satisfied under identical concave copulas ($C = C^{IM} = C^{MM}$).

With re-sorting, since $\int_0^1 C^{IM}_2(u,v)\,dv = \int_0^1 C^{MM}_2(u,v)\,dv = u$, the difference $C^{IM}_2(u,\cdot) - C^{MM}_2(u,\cdot)$ must change sign, so the ordering of $C^{IM}_2(u,a)$ and $C^{MM}_2(u,b)$ is not determined for arbitrary $a \geq b$, and the sign of $MC^{IM}(p) - MC^{MM}(p)$ is ambiguous. □

To recover a marginal cost-slope definition of increased adverse selection, the notion of re-sorting must be strengthened. Specifically, it requires additionally satisfies a monotone regression dependence ordering.[6] This condition is only needed to order

[6] Specifically, $\tilde{g}^{IM}_u(g^{MM}_u(v))$ is increasing in u for all $v \in (0,1)$, where $g_x(y) = \partial C(x,y)/\partial x$ and $\tilde{g}_x \equiv g_x^{-1}$. Under normality this reduces to Assumption 4.

the *slope* of the marginal cost curve; the equilibrium results of Section 2.3 require only the weaker conditions on average costs established above.

PROPOSITION A2 (Increased Adverse Selection). *Consumers are adversely selected in both mature and immature markets. If resorting between markets satisfies the following additional assumption:*

i. *[Monotone Regression Dependence Ordering]* $\tilde{g}_u^{IM}\left(g_u^{MM}(v)\right)$ *is increasing in* u *for all* $v \in (0,1)$*, where* $g_x(y) \equiv \frac{\partial C(x,y)}{\partial x}$ *is the conditional distribution of* V *given* $U = x$*, and* $\tilde{g}_x \equiv g_x^{-1}$ *is its generalized inverse.*

Then consumers are more adversely selected in immature markets $\left(\frac{\partial MC^{IM}(p)}{\partial p} \geq \frac{\partial MC^{MM}(p)}{\partial p} > 0\right)$

PROOF OF PROPOSITION A2. The proof in the general case follows from a direct application of Proposition 2.1 in Fang and Joe (1992). □

PROPOSITION A3 (Cost Expansion). *Average costs are increasing in price and early adopters are higher cost on average,* $AC^{IM}(p) \geq AC^{MM}(p)$.

PROOF OF PROPOSITION A3. The within-market ordering of average costs established in Proposition 3 Part (ii) relies on condition (ii) of that proposition – the corner-set monotonicity of the copula. As noted in the accompanying corollary, this is the relevant primitive for the cross-market comparison here. The necessary and sufficient condition for stochastic dominance of the truncated conditional distributions (see the proof of Proposition 3 and equation (A9)) is:

$$F_{\pi|\omega\geq p}(\pi|\omega \geq v_1) \leq F_{\pi|\omega\geq p}(\pi|\omega \geq v_2)\ \forall v_1 \geq v_2 \text{ and } \forall \pi \tag{A40}$$

$$\frac{u - C(u, F_\omega(v_1))}{1 - F_\omega(v_1)} \leq \frac{u - C(u, F_\omega(v_2))}{1 - F_\omega(v_2)}\ \forall v_1 \geq v_2 \text{ and } \forall u. \tag{A41}$$

The analogous condition to order average costs between markets is:

$$F^{IM}_{\pi|\omega\geq p}(\pi|\omega \geq v) \leq F^{MM}_{\pi|\omega\geq p}(\pi|\omega \geq v)\ \forall v, \pi \tag{A42}$$

$$\frac{u - C^{IM}(u, F^{IM}_\omega(v))}{1 - F^{IM}_\omega(v)} \leq \frac{u - C^{MM}(u, F^{MM}_\omega(v))}{1 - F^{MM}_\omega(v)}\ \forall v, \forall u \tag{A43}$$

As $F_{\omega}^{MM}(\omega) \leq F_{\omega}^{IM}(\omega)$, established in Proposition A1, the following is true for all copulas satisfying condition (ii) of Proposition 3 (see the proof above)

$$\frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{MM}(v))}{1 - F_{\omega}^{MM}(v)} \;\forall v \text{ and } \forall u, \tag{A44}$$

and by the concordance ordering in Assumption 3

$$\frac{u - C^{IM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \leq \frac{u - C^{MM}(u, F_{\omega}^{IM}(v))}{1 - F_{\omega}^{IM}(v)} \;\forall v \text{ and } u \in [0,1], \tag{A45}$$

which completes the proof. □

The sufficient primitive for Proposition A3 is weaker than Assumption 3. The proof shows that $AC^{IM}(p) \geq AC^{MM}(p)$ follows directly from

$$F_{\pi|\omega \geq p}^{IM}(\pi|\,p) \leq F_{\pi|\omega \geq p}^{MM}(\pi|\,p) \quad \forall\, p, \pi, \tag{A46}$$

i.e., $\pi|(\omega^{IM} \geq p) \succeq_{FOSD} \pi|(\omega^{MM} \geq p)$ for all p — upper-tail cost dominance across markets. Assumption 3 is sufficient for this but not necessary: it delivers the condition by combining demand contraction with the within-market corner-set property, but the cross-market ordering could in principle hold without global concordance of the copulas. Concordance is retained as the maintained assumption because it also delivers Proposition A4 directly, providing a unified condition for both price-space and quantity-space comparisons.

PROPOSITION A4 (Cost Expansion Fixing Demand). *Average Costs as a function of quantity are larger in immature markets, $AC^{IM}(q) \geq AC^{MM}(q)$. Marginal Costs rotate around a point and are higher at low quantities, but smaller at high quantities, thus $MC^{IM}(q) \geq MC^{MM}(q)$ for all $q \leq \tilde{q}$ and vice versa. Absent re-sorting, both cost curves are identical.*

Moreover, the adverse selection wedge is higher at any quantity demanded: $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$.

PROOF OF PROPOSITION A4. Note that the copula defines *rank* dependence. Consequently, they are invariant under strictly monotone transformations, including the inverse demand transformation considered here. Thus, to order average costs given quantity, q, we have the following condition:

$$\frac{u - C^{IM}(u, 1-q)}{q} \leq \frac{u - C^{MM}(u, 1-q)}{q} \;\forall q \in (0,1] \text{ and } u \in [0,1], \tag{A47}$$

which simplifies to

$$C^{IM}(u, 1-q) \geq C^{MM}(u, 1-q) \; \forall q \in (0,1] \; and \; u \in [0,1]. \tag{A48}$$

This only depends on the change in sorting ($C^{IM} \neq C^{MM}$) and holds with equality under no re-sorting ($C = C^{IM} = C^{MM}$). The stated Proposition then follows from the concordance ordering in Assumption 3.

Equation (A39) gives an identical condition for marginal costs in terms of the conditional density which corresponds to stochastic dominance

$$C_2^{IM}(u, 1-q) \geq C_2^{MM}(u, 1-q) \; \forall q \in (0,1] \; and \; u \in [0,1]. \tag{A49}$$

This cannot be satisfied as stochastic dominance for all conditional distributions implies stochastic dominance of the unconditional distribution. This violates assumption 1.

Note that at $q = 0$ marginal costs are equal to average costs. Then, the monotonicity of marginal costs and the results of Proposition A2 imply a single crossing property which completes the proof of rotation.

The rotation of the marginal cost curve and Proposition A2 imply the derivative of the marginal cost curve is also ordered in quantity space

$$\frac{\partial MC^{IM}(q)}{\partial p} \leq \frac{\partial MC^{MM}(q)}{\partial q} < 0, \tag{A50}$$

and we can express the wedge as

$$W(q) = \frac{1}{q} \int_0^q (MC(t) - MC(q)) \; dt = -\int_0^q MC'(t) \frac{t}{q} \, dt. \tag{A51}$$

Combining them gives $|AC^{IM}(q) - MC^{IM}(q)| \geq |AC^{MM}(q) - MC^{MM}(q)|$. □

A.3.3. Proofs for Imperfect Competition

PROOF OF LEMMA 1. I omit the subscript j for convenience. Let $S(q)$ denote the supply schedule in a market,

$$S(q) = MC(q) + \frac{1}{\eta_D(q)} \tag{A52}$$

Excess supply is given by $ES(q) = p(q) - S(q)$ and $ES^{MM}(q^{MM}) = 0$ by definition. Then

$$ES^{IM}(q^{MM}) = p^{IM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM})\right) \tag{A53}$$

$$\leq p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM})\right), \tag{A54}$$

by the assumption of the demand contraction. Under the restriction on quantities and elasticities stated in the lemma we have

$$p^{MM}(q^{MM}) - \left(MC^{IM}(q^{MM}) + 1/\eta_D^{IM}(q^{MM})\right) \tag{A55}$$

$$= \underbrace{p^{MM}(q^{MM}) - \left(MC^{MM}(q^{MM}) + 1/\eta_D^{MM}(q^{MM})\right)}_{=ES^{MM}(q^{MM})=0} \tag{A56}$$

$$+ \underbrace{\left(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM})\right)}_{\leq 0} + \underbrace{(1/\eta_D^{MM}(q^{MM}) - 1/\eta_D^{IM}(q^{MM}))}_{\leq -(MC^{MM}(q^{MM}) - MC^{IM}(q^{MM}))} \leq 0,$$

which establishes excess supply in immature markets at the mature market equilibrium. The remainder of the proof follows analogously to the proof of Proposition 1. Thus $q_j^{MM} \geq q_j^{IM}$. It follows that $p_j^{MM} \leq p_j^{IM}$ by the rotation of the marginal cost curve and restriction on elasticities whenever the supply curve is monotone. □

Appendix B. Proofs for Section 3

Proof of the first part. In immature markets let $-C < 0$ denote the loss when both firms enter and $\Pi_M^{IM} < 0$ denote the monopolist profit, respectively. Equivalently Π_M^{MM} denotes the monopolist profit and $\Pi^{MM} = 0$ denotes the (competitive) profit in mature markets.

	enter	exit
enter	(0, 0)	(Π_M^{MM},0)
exit	(0, Π_M^{MM})	(0,0)

(a) Mature Market

	enter	exit
enter	(-C, -C)	(Π_M^{IM},0)
exit	(0, Π_M^{IM})	(0,0)

(b) Immature Market

TABLE A1. Second Period Payoff Matrix

Payoffs in the second subgame when at least one firm entered in the first period are given in Panel A of Table A1.

In a mature market that does not fully unravel $\Pi_M^{MM} > 0$ and entering is a profitable deviation for both firms if neither enter, thus this cannot be a nash equilibrium of the subgame. There is no profitable deviation from the symmetric strategy profile (enter, enter) as each firm also earns zero under unilateral deviation.

Next, consider the second subgame when neither firm entered in the first period (Panel B). When $\Pi_M^{IM} < 0$, exiting is a strictly dominant strategy and there is a unique equilibrium.

Taking these contingent strategy profiles as given, the normal form representation of the first subgame is in Table A2. As the market is immature and firms earn zero equilibrium profits in mature markets, these payoffs are identical to the immature market second period subgame and exit is strictly dominant.

	enter	exit
enter	$(-C+0, -C+0)$	$(\Pi_M^{IM}+0, 0+0)$
exit	$(0+0, \Pi_M^{IM}+0)$	$(0+0, 0+0)$

TABLE A2. First Period

Therefore, the following symmetric strategy profile is an SPE:

- First period: Play *exit*
- Second period: Play *exit* if no firm has entered, otherwise *enter*

and in equilibrium no firm enters in either period.

Proof of second part. When the market is immature in the second period, the SPE has firms playing strictly dominant strategies. Therefore, only refinements for the mature second period subgame need to be considered.

As this is a two player, two action game, the set of trembling hand perfect equilibria is equivalent to the set of equilibria in which no player plays a weakly dominated strategy. The equilibria considered above (enter, enter) when mature satisfies this. However, the asymmetric strategy profiles in the second round, either (entry, exit) or (exit, entry), do not because entering weakly dominates exiting.

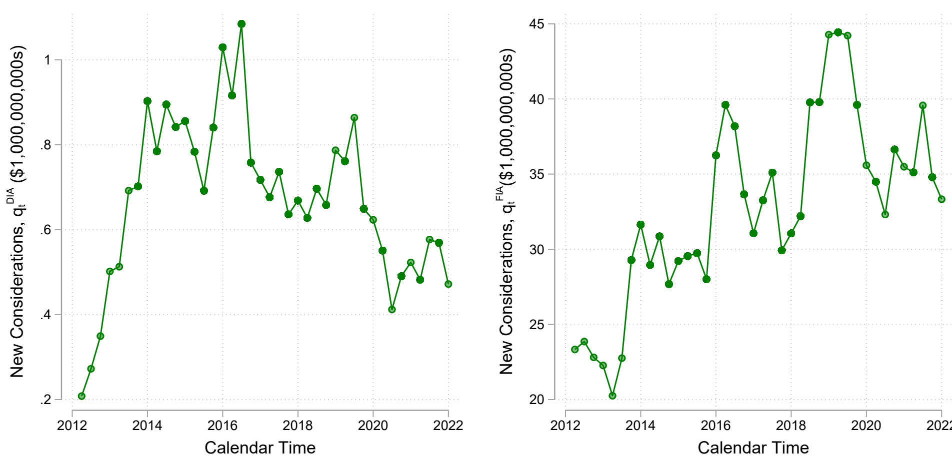
This leaves a unique equilibria surviving the trembling-hand refinement in this subgame. The same argument can be applied in the same fashion to the first period delivering a unique equilibria of the extensive form game.

Appendix C. Additional Discussion of Section 4

Testimonials. To highlight the widespread use of the Annuity Shopper quotes I reproduced the following testimonial from the July 2005 issue. The testimonial is addressed to Hersh Stein (the publisher):

> Dear Hersh:
>
> Thank you for providing the WebAnnuties packet and the Annuity Shopper. It is the best source of information I have found on the subject of annuities. And a special thanks for calling and answering my questions.
>
> As an attorney who frequently represents clients regarding investments, retirement and estate planning I will highly recommend you for their annuity needs. As I mentioned I have several annuities now and will continue to add more to my own retirement portfolio.
>
> I have recommended you and your company to other estate planners and financial advisors.
>
> Very truly yours, David D. Dunakey

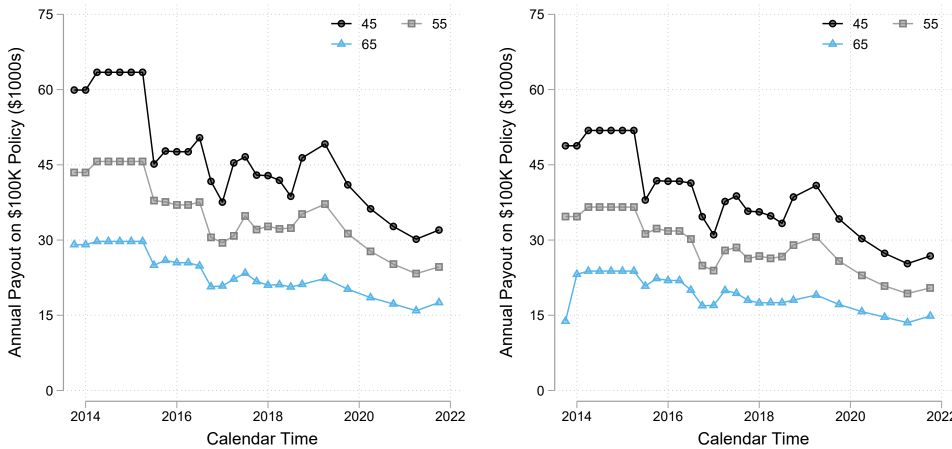
Similar testimonies are frequently published.



(A) Deferred Income Annuities (B) Other Fixed Income Annuities

FIGURE A2. Growth In The Deferred Income Annuities Market

Sales of Annuity Products. Figure A2 shows how the size of this market has evolved over time with the left panel displaying new deferred income annuity sales. At the start of this period the market for deferred income annuities was small, totaling around $200 million and accounting for less than 1% of the overall fixed income annuity market. Subsequently, however, the market expanded rapidly between 2012 and 2014, with the real value of new considerations more than quadrupling in size. The right panel shows sales of all other fixed income annuity products, the most similar alternative product in the annuity space. Comparing the two panels suggests an element of common cyclicality, but growth in fixed income annuities is much smaller over the period.



(A) New York Life (Men) (B) New York Life (Women)

FIGURE A3. Average Payouts from Deferred Income Annuity Policies Beginning Payment at Age 80 By Gender and Purchase Age

Within Insurer Payout Variation. Figure A3 repeats the exercise in Figure 5 for contracts offered by New York Life, but conditioning on the deferral age and varying the purchase age. The left panel shows contracts for men and the right panel for women with each line corresponding to a different purchase age between 45 and 65. Again, contracts that offer payouts further into the future have higher payout amounts. Similarly, payouts for men who have lower expected longevity at all ages are larger than for women. Despite systematic differences by product characteristics, a notable feature of both Figures 5 and A3 is evidence of a market level trend in payouts over time for all contract terms and for both insurers. This reflects that the largest cost to the insurer is the opportunity cost of foregone investment (see Mitchell et al. 1999) as well as a changing pool of annuitants.

C.1. Additional Empirical Results for Section 4

		Alternative Controls		Alternative Selection Measures			
	(1) Base	(2) Time FEs	(3) Policy FEs	(4) Levels	(5) IAM Life Table	(6) 10-Year Treasury	(7) Treasury Yield Curve
$\ln Q_t$	0.287^{***}		0.284^{***}	0.052^{***}	0.067^{***}	0.065^{***}	-0.006
	(0.030)		(0.030)	(0.005)	(0.008)	(0.008)	(0.005)
$\mathbb{1}[a > 45] \ln Q_t$	-0.233^{***}	-0.256^{***}	-0.230^{***}	-0.046^{***}	-0.064^{***}	-0.043^{***}	-0.001
	(0.031)	(0.026)	(0.031)	(0.005)	(0.008)	(0.008)	(0.005)
N	5,831	5,831	5,831	5,882	5,882	5,882	5,882
R^2	0.484	0.687	0.491	0.633	0.395	0.592	0.586

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient δ in equation (22). Implied selection is recovered using equation (21) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and aggregate demand for all other FIA products (excluding (2) due to colinearity with the time specific fixed effects).

TABLE A3. Adverse Selection and Market Maturity

Alternative Controls. Table A3 presents alternative specifications corresponding to the results in Table 1. The first column repeats the baseline estimate from the final specification with the full set of controls in Table 1. The second column allows for a non-parametric time trend by replacing the control for contemporaneous demand in all other fixed income annuity markets with a series of quarterly indicators (this necessitates dropping the un-interacted $\ln Q_t$ from the controls). The parameter estimate is almost unchanged, highlighting that identification does not rely on the parametric control for aggregate market conditions. This non-parametrically absorbs all time-varying common factors, including any markup dynamics induced by firm entry, showing that entry and markup dynamics are not the principle driver of the estimated effects.

Finally, the third column replaces the controls for policy characteristics with a finer set of interact-characteristic fixed effects. Each annuity policy is identified by an insurer-gender-deferral age specific identifier and this specification controls for fixed effects along this interacted dimension. This allows for more granular differences in how insurers market to or administer plans by gender and deferral age which may drive estimates of the effect of maturity due to differential entry and exit of insurers and policy combinations. The point estimate is almost identical to the baseline specification.

Alternative Selection and Mortality Tables. The fourth column replaces the dependent variable with selection measured as deviations in levels rather than logs. The magnitude of the estimate changes due to the alternative specification of the dependent variable, but the qualitative patterns remain. The fifth column uses life tables from the individual Annuitant Mortality (IAM) 2012 table, which is described in American Academy of Actuaries / Society of Actuaries (2011). As discussed by Poterba and Solomon (2021) this forecasts significant mortality improvement at odds with observed mortality improvements. Consequently, the overall money's worth measure of annuities is more positive and the log specification discards a large share of the observations where the implied NPV exceeds the cost to consumers. Nevertheless, results across columns 4 and 5, both using levels, suggest that there are similar qualitative findings whether deviations are measured from the pool of all potential consumers or the pool of typical annuitants.

Alternative Opportunity Cost of Capital. Finally, columns six and seven consider alternative assumptions for the insurer opportunity cost of capital. Contracts with different deferral-age and purchase-age combinations have different payout horizons, so if the assumed discount rate differs markedly from the insurer's actual cost of capital this may generate spurious variation across contract types. As with alternative life tables, I present results for the level of the outcome variable. In column six, the contract-date specific interest rate uses the 10-year zero-coupon Treasury yield from the Gürkaynak et al. (2007) fitted yield curve in place of the BBB corporate bond yield. This follows recent Department of Labor recommendations for defined contribution plan sponsors (see Poterba and Solomon 2021). Despite the substantially different level and dynamics of this rate due to the underlying risk premium, the results remain statistically significant and qualitatively consistent with the baseline, confirming that the findings are not driven by the specific choice of corporate yield or the embedded credit spread. Moreover, they are very close to results in levels for the baseline selection measure which uses an analogous specification.

Horizon-Specific Discounting. Column seven discounts each cash-flow horizon at the corresponding zero-coupon rate from the full Gürkaynak et al. (2007) yield curve rather than applying a single rate to all horizons. This specification produces estimates close to zero. Since long-term risk-free rates exhibit little time-series variation, horizon-specific discounting compresses the tails of the money's worth distribution and flattens its time-series dynamics, leaving insufficient variation to identify the differential dynamics across purchase ages. This does not contradict the selection interpretation—the direct evidence from subjective mortality beliefs (Table 3) confirms that private information increases with age independently of any discount rate assumption—but it illustrates a limitation of the money's worth framework when the discount rate structure removes the cross-contract variation that the test exploits. The baseline BBB specification, which reflects an opportunity cost relevant to insurers' actual pricing decisions, preserves this variation.

Disaggregated Results. Table A4 repeats this exercise for the results reported in Table 2 which allow for heterogeneous effects by purchase age. Once again the results are robust to alternative controls and definitions of the dependent variable. Point estimates for specifications with alternative controls are almost identical and alternative specifications of the dependent variable yield similar qualitative findings that echo results reported in Table 2. This shows that results are not driven by particular specification choices or by assumptions about parameters used to calculate the net present value of insurer payouts. With the exception of column seven, all specifications in both Tables A3 and A4 the estimated effects remain economically significant and statistically significant at the 1% level.

		Alternative Controls		Alternative Selection Measures			
	(1) Base	(2) Time FEs	(3) Policy FEs	(4) Levels	(5) IAM Life Table	(6) 10-Year Treasury	(7) Treasury Yield Curve
$\ln Q_t$	0.287***		0.285***	0.052***	0.067***	0.065***	−0.006
	(0.030)		(0.030)	(0.005)	(0.008)	(0.008)	(0.005)
$\mathbb{1}[a=55]\ln Q_t$	−0.144***	−0.164***	−0.142***	−0.031***	−0.041***	−0.027***	0.002
	(0.033)	(0.027)	(0.033)	(0.006)	(0.009)	(0.009)	(0.006)
$\mathbb{1}[a=65]\ln Q_t$	−0.346***	−0.373***	−0.345***	−0.066***	−0.093***	−0.064***	−0.005
	(0.031)	(0.027)	(0.031)	(0.006)	(0.010)	(0.009)	(0.006)
N	5, 831	5, 831	5, 831	5, 882	5, 882	5, 882	5, 882
R^2	0.491	0.695	0.498	0.636	0.399	0.594	0.586

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Huber-White heteroskedasticity robust standard errors are reported. Estimates reported for coefficient $\delta_5 5$ and $\delta_6 5$ in equation (23). Implied selection is recovered using equation (21) and panel data collected from policy quotes printed in *Annuity Shopper* magazine. Aggregate quantities correspond to the cumulative real value of new deferred income annuity considerations collected from LIMRA and new considerations for all other fixed income annuity (FIA) products reported in Figure 4. All Specifications additionally control for purchase age indicators, insurer specific indicators, gender, deferral age indicators and aggregate demand for all other FIA products (excluding (2) due to colinearity with the time specific fixed effects).

Table A4. Adverse Selection and Market Maturity by Private Information

Appendix D. Health and Retirement Survey: Data & Additional Results

I use the Health and Retirement Study (HRS), a representative panel of US households aged 50 and older surveyed biennially since 1992. The analysis draws on processed data from Foltyn and Olsson (2024), who harmonize subjective survival beliefs with demographic characteristics and mortality outcomes across waves 1992–2018.[7]

The HRS elicits survival expectations with the survey instrument: "What is the percent chance that you will live to be [X] or more?" where the target age varies with respondent age (75 for those 65 or younger, 80 for ages 66–69, and so on). I rescale responses to the unit interval. The sample is restricted to ages 50–70 and to observations where 10-year mortality status is resolved—either through observed death or sufficient follow-up. I focus on this age range as individuals younger than 50 are all spouses of HRS main respondents, therefore not representative, and the 50-70 samples closely aligns to the age groups I consider for annuity purchase age. In addition, I select those with 10-year mortality resolution to reduce the role of right-censoring in the analysis. This yields 109,316 person-wave observations.

D.1. Focal Point Answers

A potential concern is that older respondents provide less noisy answers due to mortality salience rather than possessing more private information. Average subjective mortality exhibits a "flatness" bias due to focal-point heaping at 0%, 50%, and 100% (see, e.g., Hamermesh 1985; Elder 2013; Foltyn and Olsson 2024).

Figure A4 plots focal response rates by age. The share giving focal answers is roughly constant between ages 50 and 70, fluctuating between 42% and 47% with only a small upwards trend. This rules out differential response quality as an explanation for the increasing predictive power of beliefs.

Table A5 provides a direct test. Column 1 reproduces the baseline specification from Table 3. Column 2 excludes focal responses entirely. The interaction coefficient strengthens from −0.005 to −0.007, indicating that answers without heaping carry more information. The qualitative conclusion is unchanged: the predictive power of beliefs increases with age even among respondents providing non-focal answers.

[7]The replication package is available directly from the HRS at https://hrsdata.isr.umich.edu/data-products/hrs-replication-foltyn-olsson.

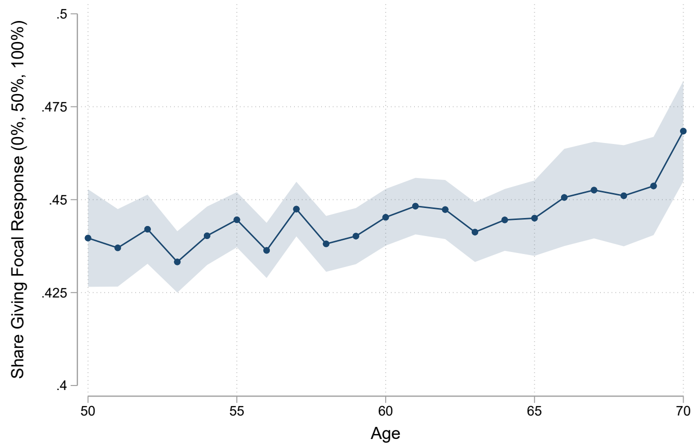


FIGURE A4. Focal Point Answers by Respondent Age
Data from the Health and Retirement Study, waves 1992–2018. Focal point answers defined as 0%, 50%, or 100%. Solid line reports sample fraction at each age pooling all cohorts and individuals. Shaded area reports 95% confidence interval.

	(1)	(2)
$\text{Belief}_{i,t}$	−0.012	0.004
	(0.011)	(0.013)
$\text{Belief}_{i,t} \times \text{Age}_{i,t}$	-0.005^{***}	-0.007^{***}
	(0.001)	(0.002)
Sample	*Full*	*Non – Focal*
N	109, 316	58, 147
R^2	0.204	0.203

Notes: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered by individual. Dependent variable is an indicator for death within 10 years. Sample restricted to respondents aged 50–70 for whom mortality outcomes are resolved estimating equation 24. Data from the Health and Retirement Study, waves 1992–2018. Column 1 reproduces analysis from Table 3 in the main text. Column 2 excludes respondents providing focal point answers of 0%, 50%, or 100%. All columns additionally control for health, gender and race.

TABLE A5. Mortality-Relevant Private Information and Age

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